Financial Foundations for Students

A Beginner’s Handbook to Saving, Investing, and Building Long-Term Wealth

Personal finance • Bank accounts and savings • Loans and eligibility • Stocks and bonds • Funds and indexes • Portfolio risk • Retirement • Options and investor protection

Audience: Students and first-time learners with no prior finance background

Context: United States financial system and regulations

Purpose: Education only - not personalized investment, tax or legal advice

An independent educational consolidation drawing on Investor.gov, FINRA, Nasdaq, OCC’s Options Industry Council, FDIC, CFPB, Federal Student Aid, HUD, VA, USDA, SBA and TreasuryDirect. Brokerage documentation explains quote fields and borrowing terminology.

How to Use This Handbook

Begin with money management, then move through saving, banking, borrowing, ownership, pooled investments, portfolios and retirement. Chapter 3A applies interest concepts to bank deposits. Chapter 3B compares bank loans, eligibility and investment securities. Chapters 4-6 explain stocks, bonds and funds in greater depth. Chapter 11 introduces options and the five Greeks alongside complex-product risks and investor protection. Practice questions, a shared answer key and an alphabetical glossary support learning.

Important Disclaimer

Investments can lose value, including all principal. Hypothetical examples are teaching illustrations, not forecasts or recommendations. Options and short selling can create substantial losses; some short positions have theoretically unlimited risk. FDIC coverage concerns eligible insured deposits; SIPC protection concerns eligible missing customer assets after broker failure, not market losses. Verify current rules and product terms with the responsible primary source. This is not an official publication of any cited organization.

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Learning Objectives

1. Define income, expenses, assets, liabilities, net worth and budgets; build a simple cash-flow plan.

2. Calculate percentages, simple and compound growth, inflation and basic investment returns.

3. Distinguish saving from investing and connect liquidity and time horizon to a financial goal.

4. Explain stock ownership and bond lending; interpret market cap, P/E, dividend yield, trading volume, session prices, 52-week ranges, short inventory and borrow rates.

5. Explain mutual funds, ETFs, NAV, distributions, expense ratios and trading differences.

6. Distinguish indexes from investable products and compare active and passive management.

7. Explain how Nasdaq's exchange, Composite and Nasdaq-100 differ.

8. Apply risk tolerance, risk capacity, diversification, concentration, asset allocation and rebalancing.

9. Read brokerage information and prospectuses; distinguish order types, fees and basic tax concepts.

10. Explain employer retirement plans, traditional and Roth treatment, retirement income and longevity risk.

11. Distinguish calls and puts, holders and writers, premium and strike, intrinsic and extrinsic value, exercise and assignment; calculate simple expiration payoffs.

12. Interpret Delta, Gamma, Theta, Vega and Rho with their units, contract scaling and model limitations.

13. Recognize complex-product risks, fraud warning signs and professional-verification resources.

14. Compare bank account types, evaluate APY and fees, calculate periodic-deposit savings, explain CD ladders and identify the scope of FDIC deposit insurance.

15. Compare bank loan eligibility and borrowing costs, distinguish federal and private student loans, identify mortgage-program conditions, and explain bank-affiliated securities access and collateral risks.

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Find interest topics: Simple Interest versus Compound Interest (Section 2.2, page 9); Simple Interest in a Banking Context (Section 3A.1, page 14); Compound Interest and Annual Percentage Yield (Section 3A.2, page 14). Chapter 3A also covers recurring deposits, checking, savings, high-yield savings, money market deposit accounts, CDs, CD ladders and deposit insurance. Chapter 3B covers loan eligibility, borrowing costs and bank-affiliated securities.

Introduction

Money is a tool. Learning how to use that tool well is one of the most practical skills a student can develop. This handbook is written for learners who have never taken a formal finance course. It begins with the everyday realities of income, expenses, and goals, then gradually introduces the concepts that allow money to work over long periods of time.

The sequence is deliberate. Before discussing stocks or funds, we establish a clear picture of personal cash flow and net worth. Before examining risk and portfolios, we explain what ownership and lending actually mean. Options are introduced after the core investing chapters, with contract mechanics and worked examples before the Greeks. Other complex products remain at an awareness level. Understanding a product does not mean it is appropriate for a beginner's portfolio.

Throughout the handbook you will find three recurring elements:

• Worked examples that use realistic student numbers and clearly labeled assumptions.

• Common misconceptions that often trip up first-time learners.

• Short practice questions at the end of each chapter so you can check understanding. The handbook draws on official educational materials from Investor.gov (the SEC's investor education site), FINRA, Nasdaq and OCC's Options Industry Council, with selected brokerage documentation for platform terminology. Hypothetical calculations are invented solely for teaching and are never presented as forecasts. Where rules change (tax limits, contribution ceilings, specific index methodologies), the text points you to the current official source rather than locking in a number that may become outdated.

This is financial education, not personalized advice. Your own goals, time horizon, tax situation, and tolerance for fluctuation will differ from those of any example character. Use the concepts here as a foundation, then verify current details and, when appropriate, consult a qualified professional.

A note on insurance protections appears repeatedly because the distinction is fundamental. Money in a deposit account at an FDIC-insured bank is protected against the failure of that bank up to the applicable limit ($250,000 per depositor, per insured bank, for each account ownership category). Securities held in a brokerage account may be protected by the Securities Investor Protection Corporation (SIPC) against the failure of the brokerage firm, subject to SIPC's limits. Neither FDIC nor SIPC protects you against a decline in the market value of stocks, bonds, mutual funds, or ETFs. Understanding this difference prevents a common and costly misunderstanding.

Chapter 1 - Financial Foundations

Source grounding: Investor.gov “Introduction to Investing,” “Figure Out Your Finances,” and related save-and-invest guidance; FDIC deposit-insurance materials.

1.1 Income, Expenses, and Cash Flow

Income is money you receive. For most students the largest sources are wages from a part-time or summer job, scholarships or grants that do not have to be repaid, gifts or support from family, and other non-borrowed support. Student-loan proceeds provide cash but create debt; they are borrowed funds rather than earned income. Some students also receive work-study funds or small side-gig payments. Expenses are the money that leaves your accounts: tuition and fees, rent or residence-hall charges, food, transportation, phone and internet plans, textbooks, entertainment, clothing, and any debt payments already under way.

The difference between income and expenses in a given period is your cash flow. Positive cash flow means you have money left after covering necessary costs; negative cash flow means you are spending more than you receive and must draw on savings or borrow. Tracking cash flow for even one full month reveals patterns that are invisible when you simply glance at the balance in a checking account. Many students discover that small, repeated purchases-coffee, rideshares, subscription services-add up to a surprisingly large monthly total.

Cash-flow tracking does not require expensive software. A simple notebook, a free spreadsheet, or the notes app on a phone works. Record every inflow and outflow for 30 days. At the end of the month, sort the outflows into needs, wants, and debt or savings contributions. The resulting picture is the starting point for every later decision in this handbook.

1.2 Building a Simple Budget

A budget is a written plan that matches expected income to planned expenses and savings. It does not have to be complicated or rigid. Many beginners start with three broad categories:

• Needs - housing, basic food, essential transportation, minimum debt payments, basic utilities and phone service required for school or work.

• Wants - dining out, streaming services, new clothes beyond essentials, entertainment, hobbies, gifts.

• Savings and debt reduction - emergency-fund contributions, retirement-account contributions, extra payments on high-interest debt. A widely cited starting guideline is the 50/30/20 framework: roughly 50 percent of after-tax income for needs, 30 percent for wants, and 20 percent for savings and debt payoff. This is a guideline, not a rule. A student living in a high-cost city may have a higher needs percentage; a student living at home with strong family support may be able to allocate far more than 20 percent to savings. The value of the framework is that it forces an intentional conversation about trade-offs.

Budgets fail most often for two reasons: they are unrealistically tight, or they are never reviewed. Build in a small buffer for unplanned but normal expenses. Review the budget at least monthly and adjust categories rather than abandoning the plan the first time reality differs from the forecast.

Hypothetical Student Budget Example - Maya Maya is a second-year college student working 12 hours per week at $15 per hour after taxes (approximately $720 per month) and receives a $400 monthly contribution

from family for living expenses. Her total monthly income is therefore about $1,120. After tracking for a month she constructs the following plan:

Category	Amount	Type	Notes
Rent share	$450	Need	Shared apartment

Category	Amount	Type	Notes
Groceries & household	$180	Need	Cooks most meals
Transit + occasional rides	$70	Need	Bus pass + rare rideshare
Phone plan	$40	Need	Family plan share
Minimum student-loan interest	$25	Need	Interest-only while in school
Dining out / coffee	$90	Want	Reduced from $140 tracked
Streaming & entertainment	$25	Want	Shared subscriptions
Clothing / personal care	$40	Want	Averaged over year
Emergency fund contribution	$100	Savings	Automatic transfer
Remaining flexible buffer	$100	Flexible	Unexpected small costs

Hypothetical example for illustration only. Actual student budgets vary widely by location, family support, and personal choices.

Notice that Maya deliberately reduced dining-out spending after seeing the tracked total. She also automated the emergency-fund transfer so that the money leaves her checking account before she can spend it. Automation is one of the most reliable behavioral tools available to beginning savers.

1.3 Assets, Liabilities, and Net Worth

An asset is something you own that has economic value. For a student this might include cash in a checking or savings account, a used car, a laptop, textbooks that can be resold, or shares of stock received as a gift. A liability is something you owe: a student-loan balance, a credit-card balance, a car loan, or money borrowed from family that must be repaid. Net worth is simply total assets minus total liabilities.

For many students, net worth is small or even negative because education debt can exceed early-career savings and the value of personal property. That is normal and expected. The purpose of calculating net worth is not to produce a score that feels discouraging; it is to create a clear baseline so progress can be measured over time. Raising net worth can happen in two ways: increasing assets (saving and investing) or reducing liabilities (paying down debt). Both directions improve the number.

Hypothetical Net-Worth Snapshot - Jordan Jordan is 20 and has the following approximate figures:

Item	Amount	Type
Checking account	$850	Asset
Savings / emergency fund	$1,200	Asset

Item	Amount	Type
Used car (estimated resale)	$3,500	Asset
Laptop and other personal property	$800	Asset
Student loans (outstanding)	($12,000)	Liability
Credit-card balance	($450)	Liability

Item	Amount	Type
Approximate net worth	($6,100)	

Hypothetical illustration only. Vehicle values are estimates; loan balances should be confirmed with servicers.

Jordan’s negative net worth is driven almost entirely by student loans. Paying down the credit-card balance first (because of its higher interest rate) will improve the number quickly and reduce interest drag. Continuing to add to the emergency fund will increase assets. Both actions move net worth in the right direction even while the student-loan balance remains.

1.4 Financial Goals and Time Horizons

Goals give direction to a budget. Without goals, money tends to drift toward the nearest convenient purchase. Short-term goals (under roughly two years) might include building an emergency fund equal to three months of essential expenses, accumulating a used-car down payment, paying for a professional certification exam, or covering the cost of a study-abroad deposit. Medium-term goals (two to ten years) could include a house down payment, graduate-school savings, or the capital needed to start a small business. Long-term goals (ten years or more) almost always include retirement and, for some people, major future education costs for children.

The time horizon of a goal strongly influences the appropriate financial vehicle. Money that may be needed within a year or two should generally stay in accounts whose value does not fluctuate much. Money that will not be needed for a decade or longer can tolerate more short-term fluctuation in exchange for the possibility of higher long-term growth. Mixing these time horizons-using stock-market investments for next semester’s tuition, for example-is one of the most common and costly mistakes beginning investors make.

1.5 Emergency Savings

An emergency fund is cash set aside specifically for unexpected but necessary expenses-car repairs, medical bills not fully covered by insurance, a sudden loss of work hours, or an urgent trip home. Investor.gov and other official educational sources consistently recommend building this fund before taking significant investment risk in stocks or stock funds. A common target is three to six months of essential living expenses, held in a federally insured savings account or similar low-volatility vehicle so the money is available when needed without having to sell investments after a market decline.

Without an emergency fund, many people turn to high-interest credit cards or payday-style loans when surprises occur. The interest costs on those forms of credit can easily exceed any investment return that might have been earned on the same dollars. An emergency fund is therefore both a risk-management tool and a cost-avoidance tool.

How much is enough? Start with one month of essential expenses if that feels achievable, then build toward three. Essential expenses are the costs you would still have to pay if you lost your job tomorrow-rent, basic

food, minimum debt payments, utilities, insurance. Discretionary spending can be cut temporarily; the emergency fund is sized against the non-discretionary core.

1.6 Understanding Debt

Not all debt is equally harmful. Debt taken for education that raises future earning power, or a mortgage on a home that is likely to be occupied for many years, can be part of a rational long-term plan. High-interest consumer debt-especially credit-card balances carried from month to month-is usually the first priority for repayment because the interest rate often exceeds what a conservative investor can reasonably expect to earn after taxes and inflation.

When comparing debt-repayment strategies, two simple approaches are common. The avalanche method requires paying the minimum on every debt and putting every extra dollar toward the balance with the highest interest rate. The snowball method also pays minimums on everything, but directs extra payments to the smallest balance first, seeking psychological momentum from quick wins. Both approaches work; the mathematically lowest-cost path is the avalanche. Choose the method you are more likely to stick with.

Student loans deserve special mention. Federal student loans often offer income-driven repayment plans, deferment or forbearance options, and potential forgiveness pathways that private loans generally do not. Before aggressively prepaying federal student loans, a student should understand the terms of the specific loans and compare the interest rate to other uses of the same dollars (emergency fund, employer retirement match, high-interest consumer debt). There is no single correct answer for every borrower.

Common Misconceptions - Chapter 1

• “I don’t make enough to budget.” Even a modest income benefits from a written plan; the plan simply becomes tighter and more intentional.

• “Net worth only matters for rich people.” Net worth is a scoreboard for everyone. Watching it rise, even slowly, reinforces productive habits.

• “Emergency funds are a waste because inflation eats them.” Inflation does reduce purchasing power, but the alternative-high-interest debt or selling investments at a bad moment-is usually more expensive.

• “All debt is bad.” Some debt finances assets or human capital that increase future income; high-interest consumer debt is the category that most often destroys wealth.

Chapter 1 Summary

Income minus expenses equals cash flow. A budget allocates that cash flow intentionally among needs, wants, and savings or debt reduction. Assets minus liabilities equals net worth. Goals with different time horizons require different tools. An emergency fund protects against forced debt or forced investment sales. High-interest consumer debt is usually repaid before aggressive investing begins. Tracking, even imperfectly, is the foundation of every later skill in this handbook.

Practice Questions - Chapter 1

1. What is the difference between an asset and a liability? Give one example of each that a typical student might have.

2. Why is an emergency fund usually recommended before large investments in stocks or stock funds?

3. Maya’s monthly income is $1,120 and her essential expenses total $765. If she wants a three-month emergency fund covering only essentials, how much should she aim to save?

4. Explain one advantage of the debt-avalanche method and one advantage of the debt-snowball method.

5. Why might a student choose to build a one-month emergency fund before trying to reach the full three-to-six-month target?

Chapter 2 - How Money Grows

Source grounding: Investor.gov compound-growth explanations and related calculator concepts; FINRA materials on risk and return relationships.

2.1 Percentages in Everyday Finance

Almost every financial rate is expressed as a percentage: interest rates on savings and loans, investment returns, inflation, expense ratios of funds, and tax rates. A percentage is simply a fraction of 100. Five percent means 5 out of every 100, or 0.05 in decimal form. When a bank advertises a 4 percent annual percentage yield (APY) on a savings account, it means that $100 left untouched for a full year would earn roughly $4 of interest (with compounding already incorporated in the APY, assuming an unchanged rate and no fees).

Being comfortable converting between percentages and decimals is essential for the rest of this chapter. To convert a percentage to a decimal, divide by 100 (or move the decimal point two places left). To convert a decimal to a percentage, multiply by 100. A return of 0.07 is 7 percent; an expense ratio of 0.15 percent is 0.0015 in decimal form.

2.2 Simple Interest versus Compound Interest

Simple interest is calculated only on the original principal. If you lend $1,000 at 5 percent simple interest for three years, you receive $50 each year and get your $1,000 back at the end; total interest is $150. The interest does not itself earn interest.

Compound interest is calculated on the principal plus any interest already earned. Interest itself starts earning interest. This is the mathematical engine behind long-term wealth building. Investor.gov describes the effect as a snowball that gathers more snow with each rotation down a hill. The longer the snowball rolls, the larger it becomes-even if the amount of new snow added each turn is modest.

Hypothetical Compounding Illustration Assume a student invests $100 at the end of every month for 40 years and earns a constant 7 percent average annual return. (Investor.gov notes that some experts consider a 7-10 percent annual rate a useful estimate for long-term diversified investments in U.S. stocks based on historic averages; actual future returns will differ, can be negative in any period, and are not guaranteed.) Under these assumptions the future value of the stream of contributions can be estimated with the future-value-of-an-annuity formula. The account would grow to roughly $262,000. Of that total, only $48,000 came from the student’s own contributions ($100 × 12 × 40); the remaining roughly $214,000 is the result of compounding.

Changing the starting age dramatically alters the required monthly contribution needed to reach a given goal. Investor.gov illustrates that reaching $500,000 by age 65 at a 7 percent assumed average annual return requires about $127 per month if contributions begin at age 18, but more than $1,000 per month if contributions begin at age 45. Time is the most powerful variable most young people control. Starting early is not about predicting markets; it is about giving compounding more periods in which to operate.

A Smaller, Shorter Example Suppose $1,000 is invested once at 6 percent annual compound interest for 10 years, with interest added once per year. After year 1 the balance is $1,060. After year 2 it is $1,060 × 1.06 = $1,123.60. After year 3 it is approximately $1,191. After 10 years the balance is approximately $1,791. The original $1,000 produced $791 of earnings. If the same $1,000 had earned only simple interest at 6 percent,

the total interest would have been $600 and the ending balance $1,600. The extra $191 is the visible effect of compounding on a single sum over a decade.

2.3 Inflation and Purchasing Power

Inflation is the general rise in the prices of goods and services over time. Even a modest average inflation rate of 2-3 percent per year steadily reduces the purchasing power of a fixed sum of money. One hundred dollars today will not buy the same basket of groceries, transportation, or housing in twenty years.

This is why simply holding cash under a mattress, or in a non-interest-bearing checking account, is not a neutral strategy; it is a slowly losing strategy against inflation. The goal of investing for long-term goals is not merely to earn a positive nominal (stated) return, but to earn a return that exceeds inflation so that real purchasing power grows.

A rough approximation sometimes used for illustration is the “rule of 72”: dividing 72 by the annual percentage rate gives an approximate number of years for a sum to double (or, in the case of inflation, for purchasing power to be cut in half). At 3 percent inflation, purchasing power halves in roughly 24 years. At 2 percent it takes about 36 years. These are approximations, not precise forecasts, but they illustrate why long-horizon money needs to grow.

2.4 Investment Returns - What the Numbers Mean

An investment return is the gain or loss on an investment over a period, usually expressed as a percentage of the amount invested. Returns can come from price appreciation (the asset becomes worth more) or from income (interest payments or dividends). Total return combines both.

Historical averages are useful for setting realistic expectations, but they are not guarantees. Investor.gov notes that some experts consider a 7-10 percent annual rate of return a useful estimate for long-term diversified investments in U.S. stocks based on historic averages. Bonds and cash equivalents have historically produced lower average returns accompanied by less volatility. Any single year, or even any single decade, can deviate sharply from the long-term average. Students should treat historical ranges as context for planning, never as promises about the next year or the next decade.

Returns are also reduced by costs. An expense ratio of 1 percent per year means that if the underlying investments earn 7 percent before fees, the investor receives approximately 6 percent after the fee. Over decades the cumulative effect of even small fee differences becomes large. This is one reason low-cost funds receive consistent attention in official investor-education materials.

Common Misconceptions - Chapter 2

• “Compound interest only matters for rich people.” The earlier and more consistently a person contributes, the larger the compounding effect becomes-regardless of starting wealth.

• “If the market returned 10 percent last year, it will return 10 percent this year.” Annual returns fluctuate widely; averages emerge only over long periods.

• “Inflation is too small to worry about.” Over 30 years, even 2.5 percent annual inflation cuts the purchasing power of a fixed sum by more than half.

• “A higher return is always better.” Higher expected returns generally come with higher risk of loss; the appropriate return target depends on time horizon and capacity for fluctuation.

Chapter 2 Summary

Percentages convert rates into comparable numbers. Compound interest allows earnings to generate further earnings, and time multiplies the effect. Inflation steadily erodes the purchasing power of money that does not grow. Investment returns are uncertain in the short run; historical long-term averages provide context but never guarantees. Costs reduce the return the investor actually keeps.

Practice Questions - Chapter 2

1. Explain in one or two sentences why compound interest is more powerful than simple interest over long periods.

2. If inflation averages 3 percent per year, roughly how much purchasing power does $1,000 lose after 20 years? (You may use the approximation that remaining purchasing power multiplies by 1 / (1.03)^20.)

3. Why does Investor.gov emphasize starting early when illustrating the monthly savings needed to reach a retirement goal?

4. What is the difference between a nominal return and a real (inflation-adjusted) return?

5. Why do small differences in annual expense ratios matter for a student who plans to invest for 30 or 40 years?

Chapter 3 - Saving versus Investing

Source grounding: Investor.gov distinction between saving and investing; FINRA investing basics; FDIC materials on deposit insurance versus non-deposit products.

3.1 Purpose and Time Horizon

Saving and investing both involve setting money aside, but they serve different purposes and operate under different risk profiles. Saving is primarily about preservation of capital and ready access. Investing is primarily about growth of capital over time and accepts the possibility of short-term losses in exchange for the possibility of higher long-term returns.

The practical dividing line is usually the time horizon. Money that may be needed within the next one to three years is generally kept in savings vehicles whose value does not swing with stock or bond markets. Money that can remain untouched for five, ten, or more years is a candidate for investment vehicles that participate in those markets. Using the wrong tool for the time horizon-investing next semester's rent money in stocks, or keeping 40-year retirement money in a low-yield savings account-creates unnecessary risk or unnecessary erosion of purchasing power.

3.2 Liquidity

Liquidity refers to how quickly and easily an asset can be converted into cash without a significant loss of value. A checking or ordinary savings account is highly liquid: the money can usually be withdrawn the same day. A certificate of deposit may charge a penalty for early withdrawal, reducing liquidity. Shares of stock or an ETF can usually be sold on any market day, but the price received may be lower (or higher) than the price originally paid, so liquidity exists but price certainty does not. Real estate is typically far less liquid; selling a property can take weeks or months and involves transaction costs.

Emergency funds and near-term goal money need high liquidity and high price stability. Long-term investment money can accept lower liquidity and price fluctuation in exchange for higher expected growth.

3.3 Common Savings Vehicles

Bank deposits and credit-union share accounts. These are federally insured-FDIC for banks, NCUA for credit unions-up to at least $250,000 per depositor, per insured institution, for each account ownership category. They offer high liquidity and virtually no risk of principal loss from market movements. Interest rates are usually modest and may not fully offset inflation over long periods. They are the default home for emergency funds and very-near-term savings.

Certificates of deposit (CDs). A traditional fixed-rate CD is a time deposit that pays a fixed interest rate for a stated term (for example, six months, one year, or five years). Early withdrawal typically incurs a penalty that reduces the effective return. CDs can be useful when the saver knows the money will not be needed before a specific date and wants a locked-in rate.

Money market funds. These are mutual funds (or sometimes other pooled vehicles) that invest in short-term, high-quality debt instruments. They aim to maintain a stable $1.00 net asset value, although that stability is not guaranteed and has been broken in rare historical episodes. Money market funds are securities, not bank deposits; they are not FDIC-insured. They generally offer next-day or same-day liquidity and, in some

interest-rate environments, higher yields than ordinary savings accounts. They can be appropriate for some short-term goals when the investor understands the absence of deposit insurance.

3.4 When Saving Is the Right Choice

Emergency funds, near-term tuition or housing deposits, a security deposit for an apartment, the down payment on a car needed within a year, or any other sum that must be available on a known short schedule are classic savings goals. Using stock-market investments for these purposes exposes the student to the risk of having to sell after a market decline, locking in a loss precisely when the cash is required. Official educational materials consistently separate short-term savings needs from long-term investment horizons for this reason.

3.5 A Simple Decision Framework

When deciding whether a sum of money belongs in a savings vehicle or an investment vehicle, ask three questions:

1. When will I need this money? (Time horizon)

2. What happens if the value drops 20 percent just before I need it? (Capacity for loss)

3. Is the account or product protected by federal deposit insurance, or is it a security whose value can fluctuate? If the money is needed soon, if a drop in value would create hardship, or if deposit insurance is an important comfort, lean toward savings vehicles. If the horizon is long and temporary declines can be waited out, investment vehicles become candidates.

Common Misconceptions - Chapter 3

• “Saving and investing are the same thing.” They share the act of setting money aside, but differ sharply in risk, expected return, and appropriate time horizon.

• “All money at a bank is FDIC-insured.” Only deposit products are insured. Stocks, bonds, mutual funds, annuities, and other non-deposit products sold through a bank are not covered by FDIC insurance.

• “Money market funds are the same as savings accounts.” Money market funds are securities; they aim for stability but are not FDIC-insured and can, in rare cases, fail to maintain a $1.00 share price.

Chapter 3 Summary

Saving prioritizes safety and access; investing prioritizes long-term growth and accepts fluctuation. Liquidity, time horizon, and insurance protections (FDIC versus none) guide the choice of vehicle. Emergency and near-term goals belong in savings; distant goals can consider investment products. Always distinguish deposit insurance from the absence of protection against market losses.

Practice Questions - Chapter 3

1. Name two characteristics that make a bank savings account suitable for an emergency fund.

2. How does FDIC insurance differ from the protection offered by SIPC at a brokerage firm?

3. Why might a money market fund be appropriate for some short-term goals but still not be FDIC-insured?

4. What is liquidity, and why does it matter when choosing between a savings account and a stock fund for a goal two years away?

Continue to Chapter 3A for bank account comparisons, APY calculations, regular savings plans and CD ladders.

Chapter 3A - Bank Accounts and Savings Products

Source grounding: FDIC deposit-insurance guidance [B1-B3]; CFPB Truth in Savings disclosures [B4-B5] and CD guidance [B6-B7]. This chapter applies Chapters 2 and 3 to everyday banking before Chapter 4 introduces securities.

Bank accounts help students receive pay, pay bills and protect near-term savings. Choose a product around the date you need the money, its access rules, its costs and its insurance coverage. All rates and product terms in the examples below are hypothetical, not current offers. Unless stated otherwise, examples assume no fees, no taxes, unchanged rates and no withdrawals; balances are rounded only at the end. Rates/APYs, minimums, withdrawal rules, fees and product terms vary by institution. Read the current account disclosure and fee schedule before opening an account.

Learning Objectives - Chapter 3A

Explain simple interest and bank compounding; distinguish an interest rate from APY; calculate balances with regular deposits; compare checking, savings, high-yield savings, money market deposit accounts and CDs; build a basic CD ladder; identify insured deposits versus securities and compare costs and access rules.

3A.1 Simple Interest in a Banking Context

Chapter 2 introduced interest on principal. In banking, simple interest is useful for estimating earnings for a stated period when interest does not itself earn interest. Let P be principal, r the annual interest rate as a decimal and t time in years: interest I = P × r × t; ending value = P + I. A bank's actual day-count method and crediting terms may differ. Do not assume every savings account or CD uses simple interest.

Sample Scenario: Elena has $2,000 for a school payment due in six months. She wants to understand a hypothetical short-term deposit quote that pays simple interest at 4% annually.

Worked Example: With t = 6/12 = 0.5, interest = $2,000 × 0.04 × 0.5 = $40. Ending value is $2,040. The annual rate is not a six-month return of 4%; the six-month return is $40 ÷ $2,000 = 2%. If interest is paid to checking rather than retained in the deposit, it does not compound there unless the checking account itself earns interest.

3A.2 Compound Interest and Annual Percentage Yield

Chapter 2 explains why earnings can generate more earnings. Here the focus is a bank's crediting schedule. For a constant annual nominal rate r compounded m times per year, balance A = P(1 + r/m)^(mt). APY expresses a year's interest including compounding under the disclosed assumptions: APY = (1 + r/m)^m - 1 in this simplified equal-period model. Do not divide APY by 12 and call it the monthly rate. For an equivalent constant monthly rate, use (1 + APY)^(1/12) - 1. Actual bank calculations may use daily balances and different day-count conventions. [B4-B5]

Sample Scenario: Jordan compares two hypothetical accounts, one quoting a 4.8% annual nominal rate compounded monthly and another quoting 4.9% APY.

Worked Example: For $1,000 in the first account, monthly rate = 0.048/12 = 0.004. After 12 months: $1,000(1.004)^12 = $1,049.07. Its APY is about 4.907%, slightly above 4.9%. The second account would end at $1,049 if its APY remains unchanged for the year. The difference is only about $0.07; a fee could overwhelm it. APY already includes compounding, so adding another compounding adjustment to APY would double count it.

Savings APYs are commonly variable. A stated APY is not a promise that the rate will remain unchanged. Compare the same time period, then examine minimum-balance requirements, rate tiers, promotional conditions and fees. A quoted APY does not tell you your net result after every fee.

3A.3 Recurring Deposits and Regular Savings Plans

In some countries, a recurring deposit is a named installment deposit product with a fixed contribution schedule and maturity. In the United States, the usual terms are automatic savings plan, recurring transfer or regular periodic deposits into a savings account. A U.S. bank may offer a specialized club account or an add-on CD; these are separate products whose terms must be checked. A recurring transfer by itself does not lock an interest rate or create a fixed maturity. [B3]

Sample Scenario: Maya can transfer $100 from checking at the end of each month for a laptop next year. She starts with a zero savings balance and schedules transfers after payday.

Worked Example: At a hypothetical constant 3.6% nominal annual rate compounded monthly, monthly rate i = 0.036/12 = 0.003. After 12 end-of-month deposits, future value = $100 × [(1.003)^12 - 1]/0.003 = $1,220.00. Contributions are $1,200 and interest is about $20.00. The first deposit earns interest for 11 months and the last for zero months at this measurement point. Beginning-of-month deposits would yield $1,220.00 × 1.003 = about $1,223.66 using unrounded values. At zero interest, the balance would simply be $1,200.

Automation supports consistency but does not create extra income. Keep enough in checking to avoid a shortfall, confirm transfer processing times and review the plan if income changes. Regular deposits into securities are investing and may involve market loss; this example uses a bank deposit.

3A.4 Checking Accounts

Checking is designed for frequent transactions: direct deposit, bill payment, debit-card purchases, checks and ATM access. Some accounts pay interest; others do not. Compare monthly maintenance fees, fee waivers, ATM charges, overdraft terms and minimums. Available balance can differ from the displayed balance because deposits may be held or transactions may be pending. A debit card spends account money; it is not the same as a credit card. [B4, B8]

Sample Scenario: Alex needs an account for wages and rent. One hypothetical account charges $8 per month unless a qualifying direct deposit condition is met. A second has no monthly fee but charges $3 for each out-of-network ATM use.

Worked Example: If Alex does not qualify for the first waiver, its annual maintenance cost is $8 × 12 = $96. Using the second account's out-of-network ATM twice per month costs $3 × 2 × 12 = $72, excluding any ATM-owner surcharge. Using its in-network ATMs would reduce those assumed charges to $0. Compare actual behavior and all fees, not just an advertised rate. A $1,000 average balance earning 0.5% APY produces roughly $5 annually under a simplified stable-balance assumption, far less than a $96 fee.

3A.5 Traditional Savings Accounts

A traditional savings account separates money for emergencies and near-term goals from everyday spending. It earns interest under the bank's terms and may provide branch access and linked transfers. Opening minimums, minimums to earn interest or waive fees and transfer rules can differ. There is no universal bank withdrawal allowance: confirm the institution's current rules, any excess-transaction fees and transfer delays. [B1, B4]

Sample Scenario: Priya wants to keep $1,500 for emergency transportation costs in an accessible account near campus. A hypothetical savings account pays 0.5% APY and charges $4 monthly below a $300 minimum daily balance.

Worked Example: If $1,500 remains for a full year at an unchanged 0.5% APY and no fees apply, ending balance = $1,500 × 1.005 = $1,507.50. Interest is $7.50. If the fee instead applied for all 12 months, fees would be $48, exceeding that gross interest by $40.50 in this simplified comparison. Actual interest would also change if fees reduce the balance. Verify the fee-waiver measurement rather than assuming a month-end balance is enough.

3A.6 High Yield Savings Accounts

High-yield savings accounts (HYSAs) are savings deposits marketed for relatively competitive yields, often with online access. The label is not a separate FDIC insurance category or a guarantee of the highest rate. Compare APY, required balances, introductory rates, account-linking rules, access channels and service. Online access does not itself determine whether a deposit is insured; verify the legal bank and coverage. [B1-B2, B4]

Sample Scenario: Sam wants to move a $3,000 emergency fund from a hypothetical 0.5% APY savings account to a 4% APY HYSA, but needs to check how quickly external transfers arrive.

Worked Example: At unchanged rates over one year, the traditional account earns $3,000 × 0.005 = $15, while the HYSA earns $3,000 × 0.04 = $120. The gross difference is $105. If Sam expects an urgent same-day expense, keeping a small immediately accessible balance elsewhere may be useful even if it earns less. The HYSA rate may change; neither calculation is a forecast.

3A.7 Money Market Deposit Accounts

A money market deposit account (MMDA) is a bank deposit that may combine interest with limited check or debit access. Minimums, tiers and transaction features vary. It is distinct from a money market mutual fund, which is a security and is not FDIC-insured. Similar names do not make the products equivalent. [B1-B3]

Sample Scenario: Luis has $5,000 for upcoming housing costs and wants occasional check access. A hypothetical MMDA offers 3% APY with a $2,500 minimum daily balance to waive a $10 monthly fee.

Worked Example: If $5,000 remains all year, the rate is unchanged and no fee applies, interest = $5,000 × 0.03 = $150 and ending balance = $5,150. A separate practical check: withdrawing $3,000 from a $5,000 balance leaves $2,000, below the fee-waiver threshold. Luis should check whether that day's balance triggers the fee and whether checking access has limits. The product is eligible for FDIC coverage only as a deposit at an insured bank within the applicable limits.

3A.8 Certificates of Deposit

A CD is a time deposit for a specified term. Many traditional CDs fix their rate, but variable-rate, step-up and callable versions exist. Early withdrawal commonly incurs a penalty; some contracts can reduce principal if earned interest is insufficient. Read the maturity date, renewal notice, grace period, interest-payment choice and early-withdrawal provision. A renewed CD may have a different rate. Emergency money needed at an uncertain date may be better kept accessible. [B4, B6-B7]

Sample Scenario: Nia has $2,000 reserved for a payment in one year and considers a hypothetical one-year fixed CD at 4% APY. She already has separate emergency savings.

Worked Example: If held for the full year with earnings retained, the CD ends at $2,000 × 1.04 = $2,080. To illustrate a different contract's penalty, suppose its disclosed annual nominal rate is 4%, it uses simple interest on a 365-day basis and its penalty is 90 days of interest. Penalty = $2,000 × 0.04 × 90/365 = $19.73. This penalty example is not derived from the 4% APY CD and is not a universal rule. The actual amount returned on early withdrawal depends on earned interest, penalty terms and the withdrawal date.

Brokered CDs are issued by banks and sold through intermediaries. Coverage depends on the issuing bank, ownership and recordkeeping requirements, and other deposits at that bank must be counted. Selling before maturity may involve a market loss rather than a standard bank withdrawal penalty. FDIC coverage does not insure a secondary-market sale price. Beginners should understand these differences before using one. [B2]

3A.9 CD Ladders

A CD ladder divides savings across different maturity dates so portions become available at intervals. It reduces the need to commit every dollar to a single maturity but does not make an unmatured CD freely accessible. Renewing a rung exposes the saver to the rates available then; future rates may be lower. Keep immediate emergency needs outside the ladder.

Sample Scenario: Ben has $6,000 beyond his emergency fund and wants some money to become available each year rather than all at once.

Worked Example: Ben places $2,000 each in one-, two- and three-year CDs, all hypothetically paying a fixed 4% APY with interest retained. At their respective maturities, values are $2,080.00, $2,163.20 and $2,249.73. These values occur at different dates; their sum is not a same-date portfolio value. When the first rung matures after one year, Ben can spend it or open a new three-year CD at the then-available rate. If every maturing rung is renewed for three years, one rung will mature annually. Three CDs at the same bank in the same ownership category share the insurance limit; they do not each create a new $250,000 allowance. [B1-B2]

3A.10 Goal and Club Savings Accounts

Goal-based savings may use separate accounts or labeled sub-balances for tuition, travel or holidays. A club account may encourage scheduled deposits and restrict access until a payout date. A digital savings bucket may be only a bookkeeping label within one account. Labels do not establish additional insurance coverage. Check access rules and fees before assigning emergency savings to a restricted plan. [B3]

Sample Scenario: Ana wants $600 for a winter trip and considers a holiday club deposit with scheduled payout before travel.

Worked Example: Ana contributes $50 at the end of each month for 12 months. Ignoring interest and fees, $50 × 12 = $600. If a hypothetical early closing fee is $15, closing early would reduce a $400 balance to $385. Ana should compare that condition with an ordinary savings account using automatic transfers and confirm the club's payout date. The product's contract, rather than its goal label, determines access.

3A.11 Specialty CDs and Savings Features

Some banks offer no-penalty CDs, add-on CDs permitting later deposits or CDs allowing a limited rate increase. These features can help with uncertain plans or regular saving, but the label alone is not enough. A no-penalty CD may require an initial waiting period and full withdrawal; an add-on CD may limit contribution size or frequency; a rate increase may require a request. Compare the entire contract with ordinary savings. [B4, B7]

Sample Scenario: Zoe might need $1,000 in six months and compares a hypothetical no-penalty CD paying 3% APY with a traditional fixed CD paying 4% APY.

Worked Example: If each remains for a full year without fees, balances are $1,030 and $1,040. The traditional CD's gross advantage is $10. A hypothetical $20 early-withdrawal penalty could outweigh that advantage, although actual earnings at six months would depend on each contract. Zoe should confirm when no-penalty withdrawal becomes available and whether it requires closing the whole CD. Flexibility has value, but the advertised label does not define all conditions.

3A.12 Deposit Insurance and Choosing an Account

FDIC insurance protects eligible deposits if an insured bank fails, generally up to $250,000 per depositor, per insured bank, for each account ownership category. Principal plus accrued interest counts toward coverage. Checking, savings, HYSAs, MMDAs and CDs can qualify. Opening multiple accounts or using different branches of the same bank does not multiply a single owner's coverage in the same category. Use FDIC

BankFind to verify the bank and EDIE to review ownership-category coverage. Federally insured credit-union share deposits use NCUA insurance rather than FDIC insurance. [B1-B2]

Stocks, bonds, mutual funds, ETFs, money market mutual funds and annuities are not FDIC-insured deposits even when sold at a bank. Treasury securities are not FDIC-insured deposits either; their federal backing is a different protection. Deposit insurance does not promise an unchanged variable APY, prevent inflation from reducing purchasing power or eliminate fees and withdrawal penalties. A financial app is not necessarily a bank; determine which legal institution holds the deposit and whether any pass-through coverage conditions are satisfied. [B1-B2]

Sample Scenario: Taylor has $20,000 in checking, $80,000 in savings and $160,000 in CDs, all individually owned at the same insured bank, with no beneficiaries and no other ownership category involved.

Worked Example: Combined deposits = $20,000 + $80,000 + $160,000 = $260,000. Under these assumptions, $250,000 is insured and $10,000 is above the limit, before considering additional accrued interest. Buying a $5,000 money market mutual fund through the bank would not turn it into an insured deposit. Taylor should use official coverage tools and verify account ownership before assuming separate protection.

Before opening an account, compare its purpose and access channels, APY and whether it varies, minimums to open/earn interest/avoid fees, the full fee schedule, transfer delays and withdrawal conditions, and the legal bank and applicable insurance. For CDs, add maturity, penalties and renewal instructions. Keep confirmation records and review statements for errors.

Common Misconceptions - Chapter 3A

• “An annual interest rate and APY are interchangeable.” APY incorporates compounding under disclosed assumptions; a nominal rate may not.

• “Regular deposits all earn interest for the full year.” Each deposit earns only for the time it remains in the account.

• “The highest advertised APY is always the best account.” Fees, changing rates, access rules and required balances can outweigh a small yield advantage.

• “Money market accounts and money market funds are the same.” An MMDA is a deposit; a money market mutual fund is a security without FDIC insurance.

• “Every CD or savings bucket gets its own insurance limit.” Coverage aggregates eligible deposits at the same insured bank in the same ownership category.

• “A CD ladder guarantees rising returns and immediate access.” Maturity spacing offers scheduled access; future renewal rates are unknown and early access may be costly.

Chapter 3A Summary

Checking supports everyday payments. Savings accounts, HYSAs and MMDAs support goals with varying access and costs. CDs exchange some flexibility for a stated term; ladders stagger maturities. Simple interest, compounding and deposit timing explain earnings, while APY supports comparison. Match the contract to the goal and confirm insurance rather than relying on a product name.

Practice Questions - Chapter 3A

1. Calculate simple interest on $1,500 at 4% annually for nine months. What is the ending value?

2. Calculate the one-year balance and APY for $1,000 at a 6% nominal annual rate compounded monthly. Why should APY not be compounded again?

3. Deposit $50 at each month-end for 12 months at a constant monthly rate of 0.003. Find contributions, ending value and interest.

4. A checking account charges $7 monthly unless a direct-deposit condition is met. What is the annual maintenance cost if the condition is never met?

5. Compare one-year gross earnings on $2,000 at an unchanged 0.5% APY and 3% APY. Name two other terms to compare.

6. How does an MMDA differ from a money market mutual fund in product type and FDIC treatment?

7. A one-year CD pays 4% APY on $2,500. What is the maturity value if earnings remain? What documents should you check before withdrawing early or renewing?

8. Divide $3,000 equally among one-, two- and three-year CDs. Describe the initial maturity schedule and one limitation of the ladder.

9. A club savings plan receives $40 monthly for 10 months, earns no interest and charges $12 on early closing. What remains if it closes after the tenth deposit and before its scheduled payout?

10. A no-penalty CD advertisement catches your eye. Name two conditions to check before relying on immediate access.

11. One person has $35,000 checking, $90,000 savings and $140,000 CDs at one insured bank in the same single-owner category. How much is above $250,000, ignoring further interest?

12. In U.S. banking, does scheduling a recurring transfer usually guarantee a fixed rate and maturity? Explain.

Chapter 3B - Bank Loans and Securities

Source grounding: CFPB consumer borrowing guidance [L1-L7]; Federal Student Aid [L8-L9]; HUD, VA and USDA mortgage guidance [L10-L12]; SBA business lending [L13-L14]; SEC Investor.gov, TreasuryDirect, FINRA and FDIC investment guidance [L15-L20]. This chapter follows bank deposits and prepares you to distinguish borrowing from investing in the later chapters.

Banks lend money to customers and may also provide access to investments through a brokerage affiliate or another registered firm. These are different activities. With a loan, you owe money and must repay it under a contract. With a security, you invest money and accept its risks. A security such as a stock or bond is also different from collateral, the asset pledged to secure a loan. Not every bank offers every product discussed here.

All scenarios are hypothetical teaching illustrations, not offers or approval predictions. Rates, APRs, fees, loan limits, underwriting standards, available securities and program rules vary and can change. Eligibility describes who may apply and what a lender or investment firm evaluates; it does not guarantee approval or suitability. There is no single minimum credit score, income or debt-to-income ratio that qualifies everyone for all bank loans. Check the institution's current criteria and the official program rules.

Learning Objectives - Chapter 3B

Compare personal, auto, student, mortgage, home-equity, revolving, secured and business borrowing; interpret APR, amortization, debt-to-income and loan-to-value measures; identify common application evidence and program-specific eligibility; compare refinancing costs; distinguish deposits, investment securities and collateral; explain account requirements and risks for securities offered through bank brokerage services.

3B.1 Eligibility and the Cost of Borrowing

Common application checks include identity and legal capacity to enter a contract, documented income or other repayment resources, credit history, existing obligations and permitted loan purpose. Some products require collateral, a down payment or a co-signer. Lenders may request identification, address, tax identification information, pay statements, tax returns, bank statements and debt balances. Accepted documents and residency requirements vary by product and institution. A co-signer is responsible for repayment, not merely a reference. [L1-L2, L4]

Debt-to-income ratio (DTI) = required monthly debt payments ÷ gross monthly income. Loan-to-value ratio (LTV) = loan balance ÷ collateral value. For home-equity borrowing, combined LTV includes other liens. These measures inform underwriting, but satisfying an illustrative ratio is not an approval rule. A budget should also consider taxes, living costs, savings and irregular expenses.

The interest rate prices interest on the balance. Annual percentage rate (APR) is a broader annualized borrowing-cost disclosure that includes specified fees under the applicable rules; it is different from a deposit's APY. Compare APR, amount financed, term, total payments, fees and any variable-rate conditions. A lower payment obtained by extending a term may increase total interest. [L1-L3]

Sample Scenario: Maya earns $3,000 gross monthly and has $450 of required monthly debt payments. A proposed loan would add $150 monthly.

Worked Example: Proposed DTI = ($450 + $150) ÷ $3,000 = 20%. Maya must still test the payment against take-home pay and living costs. For a hypothetical $5,000 loan with a 3% origination fee deducted from proceeds, the fee is $150 and cash received is $4,850, although principal owed is $5,000. The bank's APR disclosure reflects applicable charges; adding 3 percentage points to the nominal rate is not a valid APR calculation.

For fixed, fully amortizing monthly loans in the examples below, payment M = P × i ÷ [1 - (1 + i)^(-n)], where P is principal, i is the annual nominal rate divided by 12 and n is the number of payments. Examples assume equal monthly periods, no fees, no prepayments and an unchanged rate. Displayed payments are rounded; totals use unrounded payments, so the final contractual payment may differ slightly. At a zero rate, M = P/n.

3B.2 Personal Installment Loans

A personal loan provides a lump sum repaid over a term. It may be unsecured or secured; use restrictions, origination charges, fixed or variable rates and prepayment terms differ. An unsecured loan has no pledged asset, but default can still damage credit and lead to collection. Debt consolidation changes the loan structure; it does not erase debt. [L1]

Eligibility Criteria: The lender generally reviews credit reports and scores, income, existing debts, requested amount and term. A bank relationship may be a product condition. A co-signer or collateral may be required or permitted, depending on the lender. The advertised best rate may apply only to particular applicants.

Sample Scenario: Jordan needs $5,000 for an essential expense and compares a three-year personal loan with available savings and other borrowing options.

Worked Example: At a hypothetical 12% nominal annual fixed rate, i = 0.01 and n = 36. Payment = $166.07 per month. Total payments are about $5,978.58, including $978.58 interest before fees. Jordan should add any origination charge and check whether the payment leaves room for emergency savings.

3B.3 Auto Loans

An auto loan finances a vehicle, which commonly serves as collateral. The lender can repossess the vehicle following default under applicable rules; a remaining deficiency balance may still be owed. Longer terms lower the payment but can increase interest and the time spent owing more than the vehicle is worth. [L2]

Eligibility Criteria: Lenders evaluate credit, income, debt payments, down payment, requested amount and term, plus vehicle value and product rules such as age or mileage limits. They may require title documentation and specified insurance coverage. A prequalification or dealer estimate is not a final approval.

Sample Scenario: Alex chooses a $20,000 used car, pays $2,000 down and finances $18,000 for five years. Taxes and registration are paid separately in this illustration.

Worked Example: At a hypothetical fixed 7% nominal annual rate over 60 payments, monthly payment is $356.42; total loan payments are about $21,385.29, including $3,385.29 interest. Initial LTV = $18,000 ÷ $20,000 = 90%, if the lender's value is $20,000. Insurance, maintenance, taxes and registration are additional ownership costs, so the loan payment alone is not the car budget.

3B.4 Student Loans

Private student loans may be offered by banks, credit unions and other lenders. Federal Direct loans are Department of Education loans, rather than ordinary bank loans. Compare federal aid and school assistance before private borrowing because protections and repayment choices differ. [L4, L8-L9]

Eligibility Criteria for Private Loans: Requirements commonly include an eligible school and program, the lender's enrollment criteria, school certification where applicable, permitted education costs, and sufficient credit and repayment resources for the borrower or co-signer. Citizenship or residency requirements and co-signer-release rules depend on the lender. The co-signer shares responsibility for repayment; release is not automatic.

Federal Comparison: Direct Subsidized and Unsubsidized loans generally require FAFSA completion, general federal-aid eligibility, enrollment at least half-time in an eligible program at a participating school, and available annual and aggregate eligibility. Subsidized eligibility also requires financial need and is for eligible

undergraduates. These loans generally do not use ordinary bank credit underwriting. PLUS loans have different requirements, and federal limits and eligibility changed beginning July 1, 2026; the school and current StudentAid.gov guidance should confirm the rules for the student's borrowing date and history. [L8-L9]

Sample Scenario: Priya has a $10,000 private student-loan balance when repayment begins and considers a ten-year fixed-rate repayment schedule. Her parent is considering co-signing.

Worked Example: At a hypothetical fixed 6% nominal annual rate over 120 months, payment = $111.02; total payments ≈ $13,322.46, including $3,322.46 interest. Separately, if $10,000 accrues simple interest at 6% for two years before repayment, interest = $1,200. If the contract permits that interest to capitalize, repayment principal would become $11,200, so the $111.02 illustration would no longer apply. Confirm when interest accrues, whether it capitalizes, deferment terms and the co-signer's obligations.

3B.5 Mortgages and Home Purchase Programs

A mortgage finances a home and is secured by it. Failure to pay can lead to foreclosure. A fixed-rate mortgage keeps its stated interest rate; an adjustable-rate mortgage (ARM) can change after an initial period under its index, margin and adjustment caps. Principal and interest are only part of housing costs: property taxes, homeowners insurance, mortgage insurance when required, association charges, repairs and maintenance also matter. Compare the Loan Estimate and final Closing Disclosure. [L3]

Eligibility Criteria: Lenders commonly verify income and repayment ability, credit history, existing debts, funds for down payment and closing, property value and title, occupancy and program requirements. A mortgage preapproval remains subject to conditions and the property review. There is no universal 20% down-payment rule or universal qualifying credit score.

Sample Scenario: Nia buys a hypothetical $300,000 home, pays $60,000 down and borrows $240,000 on a 30-year fixed mortgage.

Worked Example: Initial LTV = $240,000 ÷ $300,000 = 80%. At a hypothetical 6% nominal annual rate over 360 months, principal-and-interest payment ≈ $1,438.92. Adding assumed monthly property taxes of $250 and insurance of $100 gives $1,788.92 before other costs. With $600 of other required debt payments and $6,000 gross monthly income, an illustrative DTI using those housing charges is ($1,788.92 + $600) ÷ $6,000 ≈ 39.82%. This is a calculation, not a qualifying cutoff.

Conventional Mortgages

Conventional loans are not FHA-insured or VA/USDA-guaranteed. Conforming and jumbo products differ; lenders and loan programs set credit, income, down-payment, reserve and property requirements. Some conventional programs permit less than 20% down; mortgage insurance may be required.

Sample Scenario: Sam compares a hypothetical conventional program accepting 5% down with another requiring more funds upfront.

Worked Example: On a $200,000 purchase, 5% down is $10,000 and base loan principal is $190,000 before any financed charges. Sam must separately budget closing costs and any mortgage-insurance expense. The arithmetic does not establish program eligibility. [L3]

FHA Insured Mortgages

An FHA-approved lender originates the loan and FHA insures it. FHA can permit a low down payment, but insurance does not relieve the borrower of repayment. Borrowers and properties must meet current FHA requirements, and lenders can impose additional underwriting standards. Credit, repayment ability, occupancy and appraisal requirements still apply. [L10]

Sample Scenario: Elena considers an FHA program for a primary residence and wants to understand the cash required under a hypothetical 3.5% down-payment option.

Worked Example: For a $200,000 price, 3.5% = $7,000 and base principal = $193,000 before mortgage-insurance and other charges. A low down payment does not eliminate upfront or ongoing mortgage-insurance costs. Elena should confirm the credit and other requirements for that option with an FHA-approved lender; 3.5% is not available to every applicant.

VA Backed Mortgages

VA-backed home loans are available through participating lenders to eligible veterans, service members and certain surviving spouses. A Certificate of Eligibility (COE) documents service-based entitlement; the lender also evaluates credit, income, repayment capacity and the property. A COE alone is not loan approval. A funding fee may apply unless exempt, and transaction-specific down-payment conditions must be checked. [L11]

Sample Scenario: Luis has a COE and asks a participating lender whether a particular purchase can be financed without a down payment.

Worked Example: If the lender and program permit a $250,000 purchase with no down payment, base principal is $250,000 before any financed eligible fees. Luis still needs a plan for applicable closing costs and housing expenses. Verifying the COE, entitlement, lender conditions and property requirements is the practical eligibility check; having $0 down does not mean $0 cash or $0 cost.

USDA Guaranteed Mortgages

Participating lenders offer USDA-guaranteed loans for eligible households and properties in qualifying rural areas. Applicants must satisfy income, repayment, occupancy and citizenship or eligible noncitizen requirements under the program. The official property map and household-income calculation matter; a rural-looking location or modest personal salary alone does not establish eligibility. [L12]

Sample Scenario: Ana's household earns $70,000 and wants a home in a location shown as eligible on USDA's map.

Worked Example: If the applicable hypothetical household-income limit is $75,000 and the program's calculated income is $70,000, the household is $5,000 below that limit. Ana has passed only that illustrative income comparison; the lender must still verify property eligibility, income adjustments and all other requirements. The $75,000 figure is invented, not a current USDA limit.

3B.6 Home Equity Loans and HELOCs

A home-equity loan usually provides a lump sum with an installment schedule. A home-equity line of credit (HELOC) allows borrowing up to a limit during a draw period, followed by repayment under the contract. HELOC rates commonly vary, and payments can rise when rates change or principal repayment begins. Both pledge the home, so default can risk foreclosure. [L5]

Eligibility Criteria: The lender evaluates available equity, total liens, appraisal or other valuation, credit, income, debt payments and permitted property use. Maximum combined LTV and fees vary. A large amount of equity does not replace the need to show repayment capacity.

Sample Scenario: Ben's home is valued at $300,000 with a $180,000 first mortgage. A hypothetical lender permits total liens up to 80% of value.

Worked Example: Total permitted liens = $300,000 × 0.80 = $240,000; illustrative equity borrowing headroom = $240,000 - $180,000 = $60,000, before other limits or charges. If Ben takes a $20,000 fixed home-equity loan at 8% for 60 months, payment ≈ $405.53. Separately, on a HELOC with $10,000 drawn, 9%

nominal annual interest and interest-only monthly payments, a simplified month's interest is $75. At 11%, it becomes $91.67; neither payment reduces principal. Actual HELOC balance, day-count and repayment terms govern.

3B.7 Credit Cards and Personal Lines of Credit

A credit card or personal line permits revolving borrowing within a limit. Interest is generally charged on outstanding borrowing under the agreement, not simply on the unused limit. Cards may have a purchase grace period if conditions are met; cash advances often have different fees and terms. Minimum payments can leave debt outstanding for a long time. A personal line's payment and access rules may differ from a card. [L6]

Eligibility Criteria: Issuers evaluate identity, income or assets available for repayment, credit and existing obligations. For applicants under 21, card rules generally require independent ability to pay or an eligible co-signer, guarantor or joint applicant aged 21 or older who can pay; not all issuers offer those arrangements. Personal-line criteria depend on the bank.

Sample Scenario: Zoe is 19 and has earned income. She wants to compare a card with a personal line rather than assume student status guarantees approval.

Worked Example: For a hypothetical $1,000 card balance maintained for 30 days at 24% nominal annual interest on a 365-day basis, interest ≈ $1,000 × 0.24 × 30/365 = $19.73, ignoring grace periods, fees and daily balance changes. A separate personal line charging 12% on $500 drawn has simplified monthly interest of $500 × 0.12/12 = $5; a $2,000 credit limit does not mean interest is charged on $2,000. Zoe must check required principal payments and access fees for each product.

3B.8 Deposit Secured and Credit Builder Loans

A deposit-secured loan pledges a savings balance or CD; access to the pledged deposit may be restricted. A credit-builder loan commonly holds the loan funds in a restricted account until repayment conditions are met. It is intended to support a payment record, rather than provide immediate spending cash. Credit improvement is not guaranteed, and missed payments may harm credit. [L7]

Eligibility Criteria: A deposit-secured loan requires acceptable collateral and the bank's repayment and documentation checks. A credit-builder product has institution-specific income, account, membership or credit conditions. Ask which credit bureaus receive reports, how fees work and when funds become available.

Sample Scenario: Taylor considers borrowing against a $1,000 CD rather than withdrawing it early. A hypothetical lender permits borrowing 90% of the pledged deposit.

Worked Example: Maximum collateral-based amount = $1,000 × 0.90 = $900, subject to approval. If the loan rate is 7% and the deposit earns 4% APY, a rough one-year comparison on $900 is $63 borrowing interest versus $36 deposit earnings if balances stayed unchanged; loan amortization makes actual costs different. Separately, a hypothetical credit-builder plan charging $52 monthly for 12 months to release $600 costs $624 - $600 = $24, ignoring deposit interest and other fees. A $600 held balance is not cash available at the start.

3B.9 Small Business Loans and Lines of Credit

Banks may provide equipment loans, commercial mortgages, working-capital loans or business lines. SBA-guaranteed programs are offered through participating lenders; an SBA guarantee protects the lender under program conditions and is not debt forgiveness for the borrower. [L13-L14]

Eligibility Criteria: Banks examine business purpose, revenue and cash flow, operating history, business and owner credit, existing debts, collateral and any required personal guarantee. Application evidence may include business formation and ownership documents, tax returns, financial statements, projections and a use-of-funds plan. SBA 7(a) requirements include eligible U.S. business operations, small-business status,

creditworthiness and repayment capacity. Ownership and residency rules are additional gates: the SBA notice effective March 1, 2026 requires 100% direct and indirect ownership by U.S. citizens or U.S. nationals with principal residence in the U.S., its territories or possessions. Confirm the latest notice with the participating lender rather than relying on older summaries. [L14]

Sample Scenario: A graduate's repair business has $1,500 monthly cash available for debt service after operating expenses and proposes $1,000 of total monthly debt payments.

Worked Example: Illustrative debt-service coverage ratio = $1,500 ÷ $1,000 = 1.5. This calculation does not demonstrate the lender's required ratio, stable future cash flow or SBA eligibility. If sales fall and available cash drops to $900, coverage becomes 0.9 and cash is insufficient for the $1,000 payment. Separate business cash-flow evidence from a personal salary or collateral valuation.

3B.10 Refinancing and Debt Consolidation

Refinancing replaces a loan with a new one. Consolidation combines debts, but may extend repayment or change protections. Compare remaining payments on the existing debt with all payments and fees under the proposed debt; a lower rate or monthly payment alone does not prove savings. [L1, L4]

Eligibility Criteria: The new lender rechecks credit, income, debts and any collateral, loan balance, vehicle or property criteria. Student refinance lenders may require graduation or other product conditions. Moving federal student debt into a private loan gives up federal benefits and protections; it is different from federal Direct Consolidation. [L4]

Sample Scenario: Sam is offered a refinance that saves $40 per month over the same remaining term but costs $600 upfront, with otherwise comparable terms.

Worked Example: Simple fee-recovery period = $600 ÷ $40 = 15 months. If only 12 payments remain, payment savings total $480, below the $600 cost. The simple comparison ignores the time value of money and any other changes. Sam should compare complete schedules and benefits, not stop at the break-even estimate.

3B.11 Bank Brokerage Services and Investment Eligibility

Investment securities may be sold through a bank's registered brokerage affiliate or partner. Not every bank provides brokerage services, and a banking login does not establish that all balances are deposits. Identify the legal firm, account agreement, compensation and costs. Chapters 4-6 explain the underlying products; this section focuses on access and eligibility. [L15, L20]

Eligibility Criteria: A brokerage commonly asks for identity, address, tax identification, financial information, investing experience, objectives and risk information. Adults may open permitted accounts; minors generally need an authorized custodial arrangement. Product minimums, jurisdiction restrictions and account rules vary. A cash account normally requires funding for purchases; a margin account requires separate approval and borrowing conditions. Ordinary publicly traded stocks, bonds, mutual funds and ETFs do not generally require accredited-investor status merely because they are sold through a bank; private offerings can have different restrictions. [L15]

Sample Scenario: Elena sees a $2,000 savings balance and a $2,000 fund balance in the same bank-branded application.

Worked Example: She checks the statements: one identifies an FDIC-insured bank savings deposit and the other identifies fund shares held through a broker. Only the eligible deposit is covered by FDIC insurance, subject to its limits. The fund can lose value. Applicable SIPC protection concerns eligible missing customer assets when a member brokerage fails; it does not insure market returns. Shared branding does not turn the two $2,000 balances into the same product.

3B.12 Stocks Offered Through Bank Brokerage Services

Stocks represent company ownership and can produce dividends or capital gains, but can also lose all value. Chapters 4 and 8 explain valuation and portfolio risks. Bank distribution does not guarantee a stock or make it FDIC-insured. [L16, L20]

Eligibility Criteria: An eligible brokerage account, sufficient funds, trading permission and access to the security are needed. Share prices or fractional-share availability determine purchase size; minimums and fees vary. Margin or options approval is a separate matter.

Sample Scenario: Alex uses a funded cash brokerage account through a bank affiliate to buy five shares at $40 each.

Worked Example: Purchase cost = 5 × $40 = $200, excluding fees. If price falls to $32, value = $160 and unrealized loss = $40, or 20%. The account may be eligible even though the investment does not fit Alex's near-term goal. No FDIC claim reimburses this market loss.

3B.13 Corporate and Municipal Bonds

A bond is a creditor claim, not a savings deposit. Corporate and municipal issuers can default; market prices also change with rates and other risks. A bank-issued bond is a security of the bank and is different from the bank's insured CDs. Tax treatment of municipal interest depends on the bond and the investor's circumstances; do not assume all municipal income is tax-free. [L17, L20]

Eligibility Criteria: The brokerage must offer the bond and the investor must meet account and funding requirements. Minimum denominations, lot sizes, commissions or markups and available inventory vary. Some bonds or private offerings have additional restrictions. Read the issuer disclosures, maturity, call provisions and credit risks.

Sample Scenario: Nia considers a $1,000 face-value corporate bond purchased at par with a 5% annual coupon paid twice yearly.

Worked Example: Annual coupon = $1,000 × 0.05 = $50, or $25 each six months. If Nia sells for $950 after receiving one $25 coupon, simplified total result = $950 + $25 - $1,000 = -$25 before costs. Coupon rate is not the realized return, and receiving a coupon does not guarantee full principal repayment.

3B.14 Treasury Bills Notes and Bonds

Marketable U.S. Treasury securities can be bought through brokers, dealers or financial institutions, or directly through TreasuryDirect. Treasury bills generally sell at a discount or at par and pay face value at maturity; notes and bonds pay interest under their terms. They are federal securities, not FDIC-insured deposits. Selling before maturity can produce a market loss. [L18, L20]

Eligibility Criteria: Bank-channel access requires a permitted account and sufficient funds, plus the intermediary's minimum and service terms. Treasury auction bids start at $100 and use $100 increments; an intermediary may impose different transaction minimums. TreasuryDirect has its own account eligibility and operating rules.

Sample Scenario: Luis buys a hypothetical Treasury bill with $1,000 face value for $980 and holds it until maturity.

Worked Example: Maturity gain = $1,000 - $980 = $20. Holding-period return = $20 ÷ $980 ≈ 2.04%, before costs and taxes. This is not an annual yield because the holding period has not been specified. Federal backing differs from FDIC coverage, and sale proceeds before maturity may differ from face value.

3B.15 Mutual Funds

A mutual fund pools investor money in securities and issues shares priced under its fund rules. Banks may distribute funds through brokerage services. An investor buys fund shares, not an insured bank balance; this includes money market mutual funds. Chapter 5 explains NAV, expenses and distributions. [L17, L20]

Eligibility Criteria: Meet the brokerage or fund account requirements and the minimum for the relevant share class. Retirement status, distribution channel or other conditions may affect share-class access. Examine the prospectus, fees and risks rather than assuming a bank's selection suits every investor.

Sample Scenario: Priya wants to contribute $200 to a fund with a hypothetical $500 initial minimum, even though subsequent purchases may be smaller.

Worked Example: She is $500 - $200 = $300 short of that initial minimum. In a different eligible share class with no such minimum, $200 at a $20 NAV buys 10 shares before costs. If NAV later falls to $18 without distributions, value is $180 and loss is $20. Investment eligibility and investment return are separate issues.

3B.16 Exchange Traded Funds

ETFs are funds traded on an exchange at market prices. Their price can differ from NAV; expense ratios and bid-ask spreads affect results. A bank's brokerage affiliate may offer ETFs, but these securities do not have FDIC coverage. Chapter 5 supplies the detailed fund comparison. [L17, L20]

Eligibility Criteria: A permitted funded brokerage account and access to the ETF are needed. Whole-share prices or offered fractional trading determine the practical minimum. Some leveraged, inverse or other complex ETFs require additional permissions or are unavailable on a platform.

Sample Scenario: Zoe has $250 and considers an ETF quoted at $80 per whole share on a platform without fractional ETF purchases.

Worked Example: She can buy three shares for $240 before costs, leaving $10 cash. Four shares would cost $320, beyond her funds. The ETF position can lose value; the coverage of any remaining cash depends on whether it is broker cash or a disclosed deposit sweep, rather than automatically sharing the ETF's treatment.

3B.17 Structured Notes and Other Complex Securities

Some bank brokerage channels sell structured notes, whose returns depend on contractual reference assets and may involve caps, barriers, calls or conditional principal protection. They are debt obligations of an issuer, not ordinary deposits. Promised protection can depend on holding to maturity and the issuer's ability to pay. Fees, limited liquidity and complex payoff rules matter. Awareness does not mean a product is suitable for a beginner. [L19]

Eligibility Criteria: Access depends on the offering, account, minimum purchase and the firm's product review and permissions. Some private placements restrict who can invest; accredited-investor status is not a universal rule for every structured note. Read the offering documents and understand every payoff condition before committing funds.

Sample Scenario: Ben reads a hypothetical note promising half of a reference index's positive return, with a 6% maximum return, and principal repayment at maturity only if the issuer can pay.

Worked Example: If the index rises 20%, half is 10%, but the 6% cap means a $1,000 investment pays $60 of return, for $1,060 at maturity if all stated conditions are satisfied. The contract must separately explain a falling index, early sale and issuer default. The $1,060 calculation is not a deposit guarantee or an FDIC-insured payout.

3B.18 Securities Backed Credit Lines

Some banks lend against an eligible securities portfolio through a securities-backed line of credit (SBLOC). This is borrowing secured by investments, not buying a security. A falling collateral value can trigger a demand for repayment or additional assets and forced sales. Rates often vary and the line may be callable. Such non-purpose lines generally cannot be used to buy or trade securities; brokerage margin borrowing is a different arrangement. [L19]

Eligibility Criteria: The lender determines acceptable securities, account and portfolio minimums, advance rates, concentration limits and other credit conditions. The lender may change collateral eligibility or lending values. A large portfolio does not make borrowing risk-free.

Sample Scenario: Taylor has $100,000 of eligible securities and draws $40,000 under a hypothetical line permitting borrowing up to 50% of collateral value.

Worked Example: Initial borrowing capacity is $50,000. If the portfolio falls to $70,000 and the 50% rule is unchanged, capacity falls to $35,000. The $40,000 debt exceeds it by $5,000. Taylor may have to repay or add acceptable collateral on the lender's deadline; selling investments can realize losses and taxes. Actual contract terms can require action even earlier.

Common Misconceptions - Chapter 3B

• “Meeting one credit-score or DTI number guarantees a loan.” Approval depends on the complete application, collateral and program rules; thresholds differ.

• “Loan approval proves I can afford the payment.” Your budget includes costs and goals beyond a lender's underwriting measures.

• “A co-signer is just a backup reference.” A co-signer has repayment responsibility and can suffer credit harm.

• “A government-backed mortgage or SBA loan need not be repaid.” The guarantee or insurance generally supports the lender; the borrower still owes the debt.

• “A lower monthly payment always means a cheaper loan.” A longer term and fees can raise total cost.

• “Bank-issued bonds and bank-sold funds have deposit insurance.” Securities remain distinct from eligible deposits.

• “Collateral prevents loss.” Collateral can be repossessed, foreclosed upon or sold; securities collateral can lose value rapidly.

Chapter 3B Summary

Loans create repayment obligations; securities create investment exposure. Lenders evaluate repayment capacity, credit and product-specific conditions. Mortgage, student and SBA programs add distinct eligibility gates. Compare APR, fees, term and full repayment costs, then test affordability independently. Investment account eligibility does not make a security safe or appropriate. Keep deposits, securities and pledged collateral clearly separate.

Practice Questions - Chapter 3B

1. Gross monthly income is $4,000, existing required debt payments are $600 and a proposed loan adds $200. Calculate proposed DTI. Does it guarantee approval?

2. A $5,000 personal loan has a 3% origination fee deducted from proceeds. How much cash does the borrower receive, and what principal is owed?

3. A $20,000 car purchase has a $2,000 down payment. Calculate base borrowing and LTV if lender value is $20,000. Name two additional eligibility checks.

4. Name two differences between private bank student loans and federal Direct Subsidized or Unsubsidized loans. What is a co-signer's responsibility?

5. A $300,000 house has a $60,000 down payment. Find base loan and LTV. Why is principal-and-interest payment not the entire housing budget?

6. What eligibility evidence distinguishes VA and USDA mortgages from an ordinary conventional application? Does an FHA low-down-payment option remove underwriting requirements?

7. A home is worth $300,000 and has a $180,000 mortgage. Under a hypothetical 80% combined-LTV limit, calculate additional borrowing headroom before other conditions.

8. Explain the under-21 credit-card ability-to-pay issue. Calculate simplified monthly interest on a $500 line balance at 12% nominal annual interest.

9. A hypothetical credit-builder plan charges $52 for 12 months and releases $600. Find cost before deposit earnings or other fees. Is the $600 necessarily available upfront?

10. A business has $1,500 available monthly for total debt payments of $1,000. Calculate coverage. Does that alone establish SBA eligibility?

11. Refinancing saves $40 monthly for 12 months with a $600 upfront fee and otherwise comparable terms. Does it save money on this simplified comparison? What can be lost when federal student loans are privately refinanced?

12. Five shares bought at $40 fall to $32. Find value, dollar loss and percentage loss. Does bank branding provide FDIC protection?

13. A $1,000 par bond has a 5% annual coupon paid twice yearly. Find the six-month coupon. If sold for $950 after one coupon, find the simplified result.

14. A $1,000 face-value Treasury bill costs $980. Find maturity gain and holding-period return. Is that automatically its annual yield?

15. A mutual fund requires $500 initially, and the saver has $200. Find the shortfall. With $250 available and an $80 ETF whole-share price, how many shares fit before costs?

16. A hypothetical structured note pays half of a 20% index gain, capped at 6%. Find the return on $1,000 if its conditions and issuer payment obligations are met. Name one risk that remains.

17. A $40,000 SBLOC balance has a 50% lending limit. After eligible collateral falls to $70,000, find the excess debt above permitted capacity.

18. List the documents or questions you would use to compare eligibility and costs for a bank loan versus a bank-affiliated securities account.

Chapter 4 - Stocks and Bonds

Source grounding: Investor.gov and FINRA stock and bond education; stock quote and borrowing documentation. Source IDs [S1]-[S6] are listed in the shared bibliography.

4.1 Stocks: Ownership in a Company

A share of common stock represents a fractional ownership interest in a corporation. When you buy common stock, you become a part-owner. If the company prospers, the value of your ownership stake may rise and the company may distribute a portion of its profits as dividends. If the company struggles or fails, the value of the shares can fall, potentially to zero. Stockholders typically have voting rights on major corporate matters, although many individual investors hold shares in “street name” through a broker and may not exercise those votes directly.

Two primary ways investors hope to profit from stocks are capital gains (selling the shares for more than the purchase price) and dividends (cash or additional shares distributed by the company). Neither is guaranteed. Many growth-oriented companies pay little or no dividend, preferring to reinvest earnings into expansion. Dividend-paying companies may reduce or eliminate dividends if earnings decline.

Preferred stock is a separate class that usually does not carry voting rights but has priority over common stock in the payment of dividends and in the distribution of assets if the company is liquidated. Preferred stock is less common in the portfolios of beginning individual investors than common stock or stock funds.

4.2 Stock Valuation and Income Metrics

Market cap Market capitalization estimates the equity market value of a company: share price × shares outstanding. At $50 with 200 million shares outstanding, market cap is $10 billion. Outstanding shares differ from the public float, which excludes some restricted or closely held shares. Market cap is not annual revenue, cash in the bank or enterprise value. Size alone does not establish safety or fair value. [S1]

Company size and index weighting Companies are often grouped into large-cap, mid-cap and small-cap categories. Larger companies tend to be more established and widely followed; smaller companies can offer growth potential but may have greater volatility and permanent-loss risk. These labels do not guarantee performance. Market capitalization also influences weights in many major market-cap-weighted indexes, discussed in Chapter 6.

Price-earnings ratio (P/E) P/E = share price ÷ earnings per share (EPS). A $50 share and $2.50 annual EPS imply 20× earnings. Trailing P/E typically uses the last 12 months; forward P/E uses forecast earnings. Label which one you use. Zero earnings make the ratio undefined; losses often produce a negative or unavailable figure that cannot be interpreted like a positive P/E. Compare consistent periods and accounting methods. Low P/E can reflect weak prospects; high P/E can reflect growth expectations or overvaluation. [S2]

Dividend yield Dividend yield = annual dividend per share ÷ current share price × 100. A $2 annual dividend on a $50 share gives 4%. Providers may use actual trailing payments or an indicated annual amount based on a recent regular payment. Special dividends can distort comparisons. A falling share price can increase the displayed yield even while the business deteriorates. Dividends can be reduced or stopped. Yield is not total return: price changes, payments, fees and taxes all matter. [S1, S3]

Student check If the share price rises to $60, unchanged shares imply $12 billion market cap; unchanged EPS implies 24× P/E; unchanged dividends imply about 3.33% yield. The business inputs did not change, but all three displayed metrics did.

4.3 Reading Prices and Trading Activity

Open price The opening trading price for the session defined by the provider, commonly the regular session. It may be an opening auction price and need not equal yesterday's close. Overnight news can create a gap. Specify whether extended-hours trades are included. [S3]

High today and low today The highest and lowest reported trade prices so far in the specified session. These change as trading continues and are not forecasts or necessarily available execution prices. A wide daily range suggests substantial movement, but does not explain its cause. [S3]

Volume The number of shares traded during the reported period. One matched trade of 100 shares normally contributes 100 shares of volume, not 200 because it has both a buyer and seller. A share can trade multiple times. Volume does not mean unique investors, shares outstanding or the value of trades in dollars. [S3]

Average volume Mean daily share volume over a defined lookback: sum of daily volumes ÷ number of sessions. A 10-day average and a 3-month average can differ substantially; always retain the window and provider definition. An illustrative 20-session total of 60 million shares gives 3 million shares per day. Comparing a partial day with a full-day average can mislead.

52-week high and 52-week low Highest and lowest reported prices over the provider's rolling 52-week window. Verify intraday versus closing prices, session coverage and split adjustment. Neither figure predicts the next price. Being below a past high does not establish that a stock is cheap. [S3]

Liquidity is more than volume Bid is a quoted buying price; ask is a quoted selling price. Their difference is the spread. A last trade is historical and can be stale. Examine spread, quote size and timestamp as well as volume. A market order prioritizes execution; a limit order restricts price but may not execute. Providers can differ in consolidated versus venue-specific data.

4.4 Short Selling, Short Inventory and Borrow Rates

Short inventory / shortable shares On a brokerage screen this usually means the estimated shares available to borrow through that broker at that time. It is broker-specific, can change quickly and is not a reservation or execution guarantee. Confirm the platform's definition: "short inventory" is not a universal label. [S5]

Short interest is a different measure Short interest counts outstanding short positions at a reporting date. It differs from broker borrow availability and from daily short-sale trading volume. A high short-sale volume does not directly reveal how many short positions remain open. Do not substitute these fields for one another.

How a stock short works A trader borrows shares, sells them and later purchases shares to return. Selling 100 shares at $50 and buying them back at $40 produces $1,000 gross profit before borrowing costs and other expenses. Buying back at $70 instead creates a $2,000 gross loss. A stock can rise without a fixed ceiling, so losses can be theoretically unlimited. Margin requirements also apply. [S4]

Borrow rate / stock loan fee A commonly annualized rate charged for borrowing shares; it is distinct from interest on a margin cash loan. Hard-to-borrow shares may have higher fees. The rate can change while a short remains open. The fee basis, collateral valuation, day-count convention and other charges depend on the broker. [S6]

Illustrative cost calculation Assume a constant $5,000 fee basis, 12% annual rate and a 360-day convention for 10 chargeable days: $5,000 × 0.12 × 10/360 = $16.67. This is an arithmetic illustration, not a broker quote. Actual charges may use changing daily bases and rates; proceeds interest may be calculated separately. [S5, S6]

Recall, buy-in and dividends Borrowed shares can be recalled; the broker may close the short. Availability can disappear when it is hardest to exit. A short seller may owe payments corresponding to distributions on borrowed shares. A low fee today does not eliminate tomorrow's price, funding or recall risk.

4.5 Worked Example: Reading a Stock Quote

Fictional quote - not live market data LEARN Co.; regular-session USD quote, hypothetical timestamp. Last $50; open $49; high today $51; low today $48; volume 2 million shares; average daily volume 3 million over 20 sessions; 52-week high $65; 52-week low $35. Shares outstanding 200 million; annual EPS $2.50; annual indicated dividend $2. Broker short inventory 25,000 shares; indicated borrow rate 12% annually.

Read the numbers together Market cap is $10 billion. P/E is 20× on the stated EPS basis. Indicated dividend yield is 4%. Today's range so far is $3. The opening gap cannot be calculated without the prior close. The 52-week range is historical, while today's high and low cover only the stated session.

What remains unknown The quote alone does not establish profitability trends, debt capacity, revenue quality, valuation relative to peers, dividend sustainability or suitability for a particular student. Two million shares traded so far is not necessarily unusually low: the session may be incomplete. Borrow availability at one broker is not total market short interest.

Data definitions matter

For comparisons, record units, quote timestamps, EPS period, trailing or forward P/E, dividend basis, average-volume window and price-adjustment method. An unavailable value is not zero. Historical examples cannot supply current quotes or broker borrow fees.

4.6 Bonds: Lending Money

A bond is a debt instrument. When you buy a bond, you are lending money to the issuer-a corporation, a municipality, or a government-for a stated period. In return, the issuer promises to pay you interest (the coupon) on a regular schedule, typically semi-annually, and to repay the face (par) value at maturity. Because bondholders are creditors rather than owners, they stand ahead of stockholders in the event of bankruptcy. That seniority is one reason high-quality bonds have historically been less volatile than stocks, although they still carry multiple forms of risk.

Bonds are often described by their issuer type (U.S. Treasury, municipal, corporate), their credit quality (investment-grade versus high-yield), and their maturity (short-term, intermediate, long-term). U.S. Treasury securities are backed by the full faith and credit of the U.S. government and are considered free of credit risk, though they still carry interest-rate and inflation risk.

4.7 Major Risks of Stocks and Bonds

Stock risks include market risk (the tendency of the overall market to decline in response to economic or geopolitical events), company-specific risk (poor management decisions, product failures, accounting problems, or loss of competitive position), and the risk that dividends may be reduced or eliminated. Individual stocks can lose most or all of their value; diversified stock portfolios reduce but do not eliminate the possibility of large temporary or permanent declines.

Bond risks include interest-rate risk (bond prices generally fall when market interest rates rise, and rise when rates fall), credit or default risk (the issuer may fail to make scheduled payments), inflation risk (fixed coupon payments buy fewer goods and services over time if inflation rises), and, for some bonds, call risk (the issuer may redeem the bond early when rates fall, forcing the investor to reinvest at lower yields). Even high-quality bonds are not risk-free.

Hypothetical Stock Illustration Suppose a student buys one share of a hypothetical company for $50. One year later the share price is $55 and the company paid a $1 dividend during the year. The total return for that year is ($5 capital gain + $1 dividend) / $50 = 12 percent. The next year the share price falls to $48 and no dividend is paid; the return for that year is -12.7 percent (($48 - $55) / $55). Over longer periods the sequence

of annual returns compounds, but any individual year can be strongly positive or negative. This variability is the practical meaning of stock-market risk for an individual holder.

Hypothetical Bond Illustration Suppose a student buys a newly issued five-year corporate bond with a $1,000 face value and a 4 percent annual coupon, paid semi-annually. The student receives $20 every six months. If

the student holds the bond to maturity and the issuer makes all payments, the student receives the coupons plus the $1,000 principal back. If market interest rates rise sharply after purchase, the market price of the bond will fall; if the student needs to sell before maturity, a capital loss is possible even though the issuer is still paying as promised. This is interest-rate risk in concrete form.

Common Misconceptions - Chapter 4

• “Stocks are only for rich people or experts.” Through mutual funds and ETFs (Chapter 5), students can own diversified baskets of stocks with very small amounts of money.

• “Bonds are risk-free.” Only certain U.S. Treasury securities are considered free of credit risk; all bonds carry at least interest-rate and inflation risk.

• “A stock that has risen a lot must be due for a fall, and vice versa.” Prices reflect available information and expectations; past price movement alone does not determine future movement.

“High yield, low P/E or a low price guarantees a bargain.” Each metric needs business context; past prices and a single ratio do not establish value.

Chapter 4 Summary

Stocks represent ownership; bonds represent lending. Market cap measures equity market size, P/E compares price with earnings, and dividend yield compares income with price. Prices and volumes describe trading over defined periods. Broker short inventory describes available borrowing, while borrow rates describe variable financing costs. These metrics complement financial statements and risk assessment; they cannot establish safety or suitability on their own.

Practice Questions - Chapter 4

1. In one or two sentences, contrast what a stockholder owns with what a bondholder owns.

2. What is market capitalization and why does it matter to an investor in a market-cap-weighted index fund?

3. Name two distinct risks that apply to bonds but are less central to the risks of holding common stocks.

4. Why can a bond lose market value even if the issuer continues to make every scheduled interest and principal payment?

5. A fictional stock is $80 with 50 million shares outstanding, annual EPS $4 and annual indicated dividends $1.60. Calculate market cap, P/E and yield.

6. Volume so far today is 800,000 shares. The 20-session daily average is 1 million. Can you conclude today’s final volume is unusually low?

7. A broker shows short inventory 10,000 shares. Does that mean total short interest is 10,000?

8. Assume a constant $8,000 borrow fee basis, 9% annual fee and a 360-day year for 5 days. Estimate cost.

Chapter 5 - Mutual Funds and ETFs

Source grounding: Investor.gov Mutual Funds and ETFs pages, Characteristics of Mutual Funds and ETFs Investor Bulletin, Index Funds page; FINRA Mutual Funds materials.

5.1 Pooled Investments

A mutual fund and an exchange-traded fund (ETF) both pool money from many investors and use that pool to buy a portfolio of securities-stocks, bonds, or a combination. Each investor owns shares of the fund, which represent a proportional claim on the underlying portfolio and on the income the portfolio generates. Professional managers (or rules-based processes in the case of index funds) select and maintain the holdings according to the fund's stated objective.

Pooling provides instant diversification that would be expensive and time-consuming for an individual student to replicate by buying individual stocks or bonds one at a time. It also provides professional administration of record-keeping, dividend collection, and year-end tax reporting. These features are among the reasons funds are the primary investment vehicle for most individual investors, including those just beginning.

5.2 Fund Categories

Funds are commonly categorized by the types of securities they hold (stock funds, bond funds, balanced or allocation funds), by geographic focus (U.S., international, global, emerging markets), by company size (large-cap, mid-cap, small-cap), by investment style (growth, value, blend), or by objective (current income, capital appreciation, target-date retirement). Target-date funds automatically shift their mix of stocks and bonds toward a more conservative allocation as a stated target year approaches; they are frequently used inside workplace retirement plans.

5.3 Net Asset Value (NAV)

Net asset value is the per-share value of the fund's assets minus its liabilities. For traditional mutual funds, shares are bought and redeemed directly with the fund (or through an intermediary) at the next-calculated NAV after the order is received. NAV is typically calculated once per day after the major U.S. markets close. ETFs also have an NAV calculated the same way, but ETF shares trade throughout the trading day on national securities exchanges at market prices that can differ-usually only slightly-from the prevailing NAV.

5.4 Trading Differences Between Mutual Funds and ETFs

Mutual-fund shares are generally purchased and redeemed at end-of-day NAV. An order placed at 10 a.m. and an order placed at 3 p.m. on the same day both receive the same closing NAV (assuming both are processed that day). ETF shares are bought and sold on exchanges during market hours at whatever price buyers and sellers agree upon at that moment. This continuous pricing is one reason some investors prefer ETFs when they want to trade at a specific time of day or use limit orders. For long-term buy-and-hold investors who contribute regularly and do not attempt to time intra-day moves, the practical difference is usually minor.

5.5 Distributions, Expense Ratios, and Transaction Costs

Funds distribute dividends and realized capital gains to shareholders. In a taxable brokerage account these distributions are generally taxable in the year they occur, even if the investor chooses to reinvest them in additional shares. (Tax-advantaged accounts such as IRAs and 401(k)s shelter the distributions from current tax.) The expense ratio is the annual fee charged by the fund, expressed as a percentage of assets under management. It covers investment management, administration, custody, and other operating costs. Even small differences in expense ratios compound into large differences in ending wealth over decades.

Transaction costs include any brokerage commissions (now often zero for many online trades of U.S.-listed ETFs and stocks), bid-ask spreads when buying or selling ETFs, and, for some mutual funds, sales loads (front-end loads charged at purchase or back-end loads charged at redemption). Investor.gov and FINRA both emphasize reading the standardized fee table near the front of the prospectus before investing. A fund that appears inexpensive may still carry costs that reduce returns; a fund that appears expensive may or may not deliver performance that justifies the cost.

5.6 Researching Funds Before Investing

Before investing in any fund, carefully read the fund's prospectus and most recent shareholder report. Key questions to answer include:

• What is the fund's investment objective, and does it match my own goal and time horizon?

• What are the principal risks disclosed in the prospectus?

• What is the expense ratio, and are there sales loads or other fees?

• How has the fund performed relative to its stated benchmark after fees? (Past performance is not a guarantee of future results.)

• What is the fund's portfolio turnover, and does it generate frequent taxable distributions? Official resources such as the FINRA Fund Analyzer and the SEC's EDGAR database provide tools for comparing costs and locating disclosure documents. Taking time to read before investing is one of the highest-return habits a beginning investor can develop.

Common Misconceptions - Chapter 5

• "All mutual funds and ETFs are basically the same." They differ in strategy, cost, tax efficiency, liquidity features, and trading mechanics.

• "A higher expense ratio means better management." Higher fees must be overcome by higher gross returns; many low-cost index funds have outperformed higher-cost active funds over long periods after fees.

• "ETFs are always better than mutual funds." For many long-term investors the differences are small; the choice often depends on account type, trading needs, and specific fund options available.

Chapter 5 Summary

Mutual funds and ETFs pool investor capital into diversified portfolios managed according to a stated objective. NAV is the underlying per-share value of the portfolio. Mutual funds trade once daily at NAV; ETFs trade continuously on exchanges at market prices. Expense ratios and other costs directly reduce the returns investors keep. The prospectus is the primary disclosure document and should be read before investing.

Practice Questions - Chapter 5

1. What does net asset value (NAV) represent?

2. Name one practical difference between buying a mutual-fund share and buying an ETF share.

3. Why does a 0.50 percent difference in annual expense ratio matter over a 30-year investment horizon?

4. Where should an investor look to find a fund's expense ratio and principal risks before investing?

Chapter 6 - Indexes and Index Funds

Source grounding: Investor.gov Index Funds page and related glossary entries.

6.1 What an Index Measures

A market index is a statistical measure of the performance of a defined group of securities intended to represent a market, a market segment, a sector, or an economy. Well-known examples include the S&P 500 (large U.S. companies), the Russell 2000 (smaller U.S. companies), and the Nasdaq indexes discussed in the next chapter. An index itself is not an investment product; it is a yardstick. Investors cannot buy shares of an index directly. They can buy funds or other products that seek to track the index's performance.

6.2 Weighting Methods

Most major equity indexes are market-capitalization weighted: companies with larger market caps have greater influence on the index's daily movement. A few indexes are price-weighted (the Dow Jones Industrial Average is the classic example), so that higher-priced stocks influence the index more than lower-priced stocks regardless of company size. Other approaches, sometimes grouped under the label "smart beta" or factor indexes, use transparent rules based on dividends, volatility, quality metrics, or other characteristics. The weighting method affects both the risk and the return characteristics of any fund that tracks the index.

6.3 Benchmarks versus Investable Products

An index often serves as a benchmark against which the performance of an actively managed fund or a personal portfolio is compared. If a fund claims to have "beaten the market," the relevant question is which index is being used as the definition of the market. A large-cap U.S. stock fund is appropriately compared to a large-cap U.S. index, not to a bond index or an international small-cap index. Clarity about the benchmark prevents misleading performance comparisons.

6.4 Active versus Passive Management

Active managers attempt to outperform a chosen benchmark by selecting securities they believe will do better than the average, or by adjusting exposure based on market forecasts. Passive (index) managers attempt to match the benchmark as closely as possible, usually by holding the same securities in approximately the same proportions (full replication) or by holding a representative sample. Because index funds generally trade less frequently and do not pay for large teams of research analysts, their expense ratios are often substantially lower than those of active funds in the same category.

Lower costs are one important reason many index funds have delivered higher net returns than the average active fund in the same category over long periods. Individual active funds can and do outperform in any given year or multi-year stretch; the challenge for an investor is identifying those funds in advance and holding them through periods of underperformance. Official educational materials present both approaches without declaring one universally superior; they do emphasize that costs matter and that past outperformance is not a reliable predictor of future outperformance.

6.5 Tracking Error and Other Index-Fund Considerations

An index fund may not match its index exactly. Tracking error can arise from fees and expenses, from sampling rather than full replication, from the timing of trades when the index changes membership, and from the use of derivatives in some funds. Investor.gov notes that index funds may also have less flexibility than active funds to react to sharp declines in securities that remain in the index. Before investing, an investor should understand how closely the fund has historically tracked its index and what methodology the fund uses.

Chapter 6 Summary

Indexes measure baskets of securities according to stated rules. Weighting methods determine which companies influence the index most. Index funds seek to track indexes at relatively low cost. Active management seeks to beat indexes but must overcome higher costs. Neither approach eliminates market risk. The prospectus and historical tracking data help investors evaluate how well a particular fund follows its stated index.

Practice Questions - Chapter 6

1. Can an investor buy shares of the S&P 500 Index itself? Explain briefly.

2. What is the main practical advantage often cited for index funds relative to actively managed funds in the same category?

3. What is tracking error, and why might an investor care about it when choosing an index fund?

Chapter 7 - Nasdaq: The Exchange and Its Indexes

Source grounding: Nasdaq official educational materials on the Nasdaq Composite and Nasdaq-100.

7.1 The Exchange versus the Indexes

Nasdaq is both a stock exchange-a marketplace where securities are listed and traded-and the publisher of several widely followed indexes. Confusing the exchange with any single index is a common beginner error. Companies list their shares for trading on the Nasdaq Stock Market. Those shares may or may not be included in particular Nasdaq-published indexes, depending on each index's eligibility rules and weighting methodology. The exchange is the venue; the indexes are statistical measures of groups of securities that trade on that venue (and, in some cases, elsewhere).

7.2 The Nasdaq Composite

The Nasdaq Composite Index includes virtually every security listed on the Nasdaq Stock Market-more than 3,000 companies spanning all sizes and sectors, including financial companies. Its purpose is to reflect the overall performance of the Nasdaq-listed market. There is no eligibility screen beyond basic listing requirements. Securities are weighted by total listed market capitalization (share price times total shares outstanding), with no float adjustment and no concentration caps that limit the influence of the largest companies. New listings typically enter the index on their second day of trading.

7.3 The Nasdaq-100

The Nasdaq-100 Index includes 100 of the largest non-financial companies listed on Nasdaq. Financial companies are deliberately excluded. The index uses a modified market-capitalization weighting methodology that includes concentration constraints so that no single company, and no small group of companies, dominates the index to an extreme degree. Eligibility rules include seasoning periods and trading-volume thresholds. The index is reconstituted annually and rebalanced quarterly. The Nasdaq-100 underpins the majority of Nasdaq-linked investment products-ETFs, futures, options, and structured products-used by investors worldwide.

7.4 Key Differences and Why They Matter to Investors

Because the Composite includes thousands of smaller companies and financial firms while the Nasdaq-100 focuses on 100 large non-financial firms, the two indexes can and do diverge. Divergence is especially visible around very large initial public offerings (when a big new listing enters the Composite immediately but may take time to enter the Nasdaq-100, and then at a constrained weight) and on days when financial stocks or small- and mid-cap stocks move differently from large-cap technology and growth stocks.

An investor watching the number labeled "NASDAQ" on a brokerage screen or financial website is usually seeing the Composite. Most popular Nasdaq-branded ETFs and related products track the Nasdaq-100, not the Composite. Knowing which index a fund actually tracks-information found in the fund's prospectus and fact sheet-is essential for understanding what the investor actually owns.

Chapter 7 Summary

Nasdaq is an exchange; the Composite and the Nasdaq-100 are separate indexes with different membership rules, different weighting methods, and different typical uses. Most investment products linked to the Nasdaq name track the Nasdaq-100. Always verify the underlying index in the fund prospectus rather than assuming that "Nasdaq" means one specific thing.

Practice Questions - Chapter 7

1. Does the Nasdaq-100 include banks and insurance companies? Why or why not?

2. Why might the performance of a popular Nasdaq-100 ETF differ from the number labeled “NASDAQ” on a financial website on a given day?

3. What is one structural reason the Composite and the Nasdaq-100 can diverge around a very large IPO?

Chapter 8 - Investment Risk and Portfolios

Source grounding: Investor.gov Managing Risk, Asset Allocation and Diversification; FINRA Risk and Asset Allocation and Diversification pages.

8.1 Risk Tolerance and Capacity for Loss

Risk tolerance is the emotional and psychological ability to accept fluctuations in portfolio value without abandoning a long-term plan. A person with high risk tolerance can watch an account decline 20 percent and still sleep at night and stay invested. A person with low risk tolerance may feel compelled to sell after a smaller decline, locking in losses.

Risk capacity is the financial ability to absorb losses without jeopardizing essential goals. A 22-year-old with stable employment, no dependents, and decades until retirement usually has high capacity even if emotional tolerance is still developing. A 60-year-old with limited savings and a planned retirement in five years has lower capacity regardless of how calm they feel about market swings. Capacity is determined by time horizon, other income sources, and the flexibility of spending goals; tolerance is determined by temperament and experience.

Both dimensions matter. An investor who has high capacity but low tolerance may still sell at the wrong time. An investor who has high tolerance but low capacity may take risks that endanger necessary living standards. Honest self-assessment of both is a prerequisite for sensible asset allocation.

8.2 Diversification and Concentration

Diversification means spreading investments across and within asset classes so that the poor performance of any single security, industry, or country has a limited impact on the whole portfolio. The familiar phrase “don’t put all your eggs in one basket” captures the idea. Concentration is the opposite: a large percentage of wealth in one stock, one industry, one country, or one asset class. Concentration can produce large gains if the concentrated bet succeeds, but it also produces large losses if it fails. Official educational materials from Investor.gov and FINRA consistently recommend diversification for most individual investors.

Diversification works best when the holdings do not all move together. Stocks of companies in different industries, bonds of different issuers and maturities, and securities from different countries often respond differently to the same economic news. Owning 20 stocks in the same industry provides less risk reduction than owning 20 stocks across many industries. Mutual funds and ETFs make broad diversification accessible with small dollar amounts.

8.3 Asset Allocation

Asset allocation is the decision of how to divide a portfolio among major asset classes-typically stocks (equities), bonds (fixed income), and cash or cash equivalents. Historical data show that these classes do not move in perfect lockstep; therefore a mix can reduce overall portfolio volatility relative to an all-stock portfolio while still participating in long-term growth. The appropriate mix depends on time horizon, risk tolerance, risk capacity, and the specific goals the money is intended to fund.

A student with a 40-year retirement horizon and high capacity for fluctuation might reasonably hold a high percentage in diversified stock funds. The same student saving for a house down payment needed in three years would ordinarily hold a much higher percentage in bonds or cash equivalents for that particular goal. Asset allocation is not a single number for an entire life; it can differ by goal and should evolve as time horizons shorten.

8.4 Rebalancing

Over time, the best-performing asset class grows to represent a larger share of the portfolio, causing the allocation to drift away from the original target. Rebalancing restores the target by selling a portion of the relative winners and buying more of the relative laggards. Rebalancing is a disciplined way to “buy low and sell high” without requiring forecasts. It can be done on a calendar schedule (for example, once a year) or whenever an asset class drifts beyond a chosen percentage band. Inside tax-advantaged accounts, rebalancing has no immediate tax cost; in taxable accounts, investors should consider the tax consequences of sales.

8.5 Regular Contributions and Dollar-Cost Averaging

Regular contributions-automatic monthly or per-paycheck investments-add a second powerful habit. They enforce a form of dollar-cost averaging: more shares are purchased when prices are lower and fewer when prices are higher, without any need to predict market direction. They also keep the investor participating through both rising and falling markets, which historically has been more important than the precise timing of any single purchase. Combining a sensible asset allocation, periodic rebalancing, and regular contributions creates a simple, repeatable process that does not depend on forecasting skill.

Common Misconceptions - Chapter 8

• “Diversification means I won’t lose money.” Diversification reduces company-specific and some sector risk; it does not eliminate market risk. A diversified stock portfolio can still decline substantially in a broad bear market.

• “I should wait for the perfect allocation before starting.” A reasonable allocation that is funded regularly almost always beats an ideal allocation that is never implemented.

• “Rebalancing is a way to increase returns.” Rebalancing is primarily a risk-control tool; any return effect is secondary and not guaranteed.

Chapter 8 Summary

Risk has both emotional (tolerance) and financial (capacity) dimensions. Diversification reduces the impact of any single holding or sector. Asset allocation sets the broad risk level of the portfolio according to goals and horizons. Rebalancing maintains the intended risk level as markets move. Regular contributions enforce discipline and reduce the importance of any single entry point. Together these practices form a durable foundation for long-term investors.

Practice Questions - Chapter 8

1. What is the difference between risk tolerance and risk capacity?

2. Why does a portfolio of 30 different stocks generally have lower company-specific risk than a portfolio of three stocks?

3. What is rebalancing, and what problem does it solve?

4. How does automatic monthly investing relate to the idea of dollar-cost averaging?

Chapter 9 - Practical Investing

Source grounding: Investor.gov investing-basics and account-related guidance; FINRA investor education on professionals and products.

9.1 Brokerage Accounts

A brokerage account is the account through which an individual buys and sells securities such as stocks, bonds, mutual funds, and ETFs. Broker-dealers that offer these accounts are regulated by the Securities and Exchange Commission and by FINRA. When choosing a firm, investors should verify registration and disciplinary history through the free search tools on Investor.gov and through FINRA's BrokerCheck. These tools are among the most practical investor-protection resources available to the public.

Accounts may be individual, joint, custodial (for minors), or trust accounts. Retirement accounts such as IRAs are also typically held at brokerage firms or mutual-fund companies and are subject to additional tax rules discussed in Chapter

10. The account agreement and the firm's fee schedule should be read before funding the account.

9.2 Basic Order Types

A market order executes as quickly as possible at the best available current market price. It prioritizes speed of execution over price certainty. A limit order executes only at a specified price or better; it prioritizes price over speed and may not execute at all if the market never reaches the limit. Stop orders become market or limit orders once a trigger price is reached; they are sometimes used to limit losses or protect gains, but they do not guarantee a specific execution price in fast-moving markets.

Beginning investors often start with market orders for highly liquid ETFs and stocks, or with simple limit orders when they want to control the maximum purchase price or minimum sale price. More complex order types introduce additional considerations and are not necessary for a long-term, regular-contribution approach.

9.3 Account Statements and Prospectuses

Brokerage account statements show current holdings, recent transactions, dividends and interest received, and overall account value. Reviewing statements periodically helps catch errors, unauthorized activity, and drift in asset allocation. The prospectus is the legal disclosure document for a mutual fund or ETF. It describes the fund's investment objectives, principal strategies, principal risks, fees and expenses, and performance. A summary prospectus is often available and is designed to be more accessible; the full prospectus remains the complete legal document. Reading at least the fee table and the principal-risks section before investing is a core habit of informed investors.

9.4 Fees and Investment Professionals

Fees reduce net returns. Common categories include fund expense ratios, advisory or management fees (often a percentage of assets under management), commissions or mark-ups on trades, account-maintenance fees, and sales loads on certain mutual funds. Over long periods, higher fees must be overcome by higher gross returns just to match the net result of a lower-cost alternative.

When working with an investment professional, understand whether the person is a fiduciary (required to act in the client's best interest) or is held to a suitability standard, and understand how the person is compensated-by

fees paid by the client, by commissions on products sold, or by some combination. Investor.gov and FINRA provide resources for checking registration and for understanding different types of professionals. Written agreements that clearly state services and costs protect both parties.

9.5 Introductory Tax Concepts for Investors

In taxable brokerage accounts, dividends and realized capital gains are generally taxable in the year they occur. Long-term capital gains (from assets held more than one year) currently receive preferential tax rates relative to ordinary income, but rates and rules are set by Congress and can change. Short-term gains are typically taxed as ordinary income. Tax-loss harvesting-selling positions at a loss to offset gains-is a technique some investors use, subject to wash-sale and other rules.

Retirement accounts (Chapter 10) offer tax advantages that can be substantial over decades: either tax-deferred growth (traditional) or tax-free qualified withdrawals (Roth). The interaction of investment returns, fees, and taxes determines the wealth an investor actually keeps. This handbook provides only conceptual awareness; tax situations are individual and professional tax advice may be appropriate for complex cases.

Chapter 9 Summary

Brokerage accounts provide regulated access to securities markets. Order types control the priority of speed versus price. Statements and prospectuses supply essential ongoing and pre-investment information. Fees matter and should be understood before committing capital. Professionals differ in standard of care and compensation structure. Taxes affect net results and vary by account type and holding period. Verification tools on Investor.gov and FINRA BrokerCheck are free and should be used.

Practice Questions - Chapter 9

1. Where can an investor check the registration and disciplinary history of a broker or investment adviser?
2. What is the primary purpose of a mutual-fund or ETF prospectus?
3. What is one difference between a market order and a limit order?
4. Why should an investor understand how an investment professional is compensated?

Chapter 10 - Retirement Foundations

Source grounding: Investor.gov retirement toolkit, employer-sponsored plans, and IRA pages.

10.1 Employer Plans and Matching Contributions

Many employers offer retirement savings plans such as 401(k) plans (common in the private sector), 403(b) plans (common in education and nonprofits), or 457(b) plans (common in government). Employees can contribute a portion of salary, often before federal income tax is calculated. Some employers match a percentage of employee contributions up to a stated limit-for example, a 50 percent match on the first 6 percent of salary contributed.

A matching contribution is effectively additional compensation. Failing to contribute enough to receive the full match is equivalent to leaving part of one's pay on the table. For most eligible workers, contributing at least enough to capture the full employer match is the highest-priority retirement step, ranking above almost every other investment decision. Contribution limits are set by law and change periodically; current limits should be verified with the plan administrator or IRS guidance for the relevant tax year.

10.2 Traditional versus Roth Accounts

Traditional (pre-tax) contributions to a 401(k) or traditional IRA reduce current taxable income. The money grows tax-deferred; withdrawals in retirement are taxed as ordinary income. Roth contributions are made with after-tax dollars (no current deduction). Qualified withdrawals in retirement-including both contributions and earnings-are tax-free. The better choice depends in part on whether the investor's tax rate is likely to be higher or lower in retirement than it is today-a comparison no one can make with certainty. Many savers use both types of accounts over a career to diversify their tax exposure.

Roth accounts also offer more flexible access to contributions (not earnings) before retirement under certain conditions, which can make them attractive to younger workers who value optionality. Specific withdrawal rules and exceptions should be confirmed with current IRS publications or a tax professional.

10.3 Retirement Income, Withdrawals, and Longevity Risk

Retirement income typically comes from a combination of Social Security, any pension income, systematic withdrawals from investment accounts, and possibly annuity payments. Longevity risk is the risk of outliving one's savings. Because no one knows their exact lifespan, planning often includes conservative withdrawal rates, diversified income sources, and, for some people, products that can provide lifetime income.

Required minimum distributions (RMDs) apply to most traditional retirement accounts after the owner reaches a certain age. The age and calculation rules have changed in recent years and may change again; current requirements should be verified with the IRS or a plan administrator. Failing to take an RMD can result in significant tax penalties.

10.4 Basic Annuity Concepts

An annuity is a contract with an insurance company that can provide a stream of payments for a stated period or for life. Immediate annuities begin payments soon after purchase; deferred annuities begin payments at a later date. Annuities can help manage longevity risk by converting a lump sum into a payment stream that cannot be outlived (for life annuities). They often carry fees, surrender charges for early withdrawals, and complexity in features and guarantees. Annuities are insurance products; they are not identical to mutual funds or ETFs and should be evaluated carefully against the investor's full financial picture, other income sources, and the strength of the issuing insurer. This handbook treats them only at an awareness level.

Chapter 10 Summary

Employer matches are valuable and should rarely be left unclaimed. Traditional and Roth accounts differ in the timing of taxation. Longevity risk is a central retirement-planning challenge. Annuities are one possible tool among several for generating lifetime income. Contribution limits, RMD ages, and specific product features change; always verify current figures with official sources.

Practice Questions - Chapter 10

1. Why is contributing enough to receive a full employer match usually recommended as a first priority?

2. In simple terms, what is the tax difference between a traditional 401(k) contribution and a Roth 401(k) contribution?

3. What is longevity risk, and why does it matter in retirement planning?

4. Why should current RMD rules be verified rather than memorized from an older guide?

Chapter 11 - Complex Products, Options and Investor Protection

Source grounding: Investor.gov and FINRA complex-product and investor-protection education; OCC's Options Industry Council contract and Greek education. Source IDs [S7]-[S15] appear in the shared bibliography.

11.1 Hedge Funds (Awareness Level)

Hedge funds pool capital from investors who meet certain financial or sophistication criteria and pursue flexible investment strategies that may include leverage (borrowing to increase exposure), short-selling, derivatives, and other techniques not commonly used by mutual funds. They are not subject to the same regulatory requirements as mutual funds and ETFs regarding daily redeemability, leverage limits, conflicts of interest, and the extent of required disclosure. Fees are typically higher than those of mutual funds or ETFs and often include both an asset-based management fee and a performance fee. Most beginning investors will not meet the eligibility criteria for hedge funds and will not need them to build a sound long-term portfolio.

11.2 Derivatives, Leveraged Funds, and Inverse Funds (Awareness Level)

Derivatives are financial contracts whose value is derived from an underlying asset, index, or rate. Options and futures are common examples. Leveraged ETFs seek to deliver a multiple (for example, 2x) of an index's daily return; inverse ETFs seek to deliver the opposite of an index's daily return. These products are generally designed for short-term trading by sophisticated investors. Over periods longer than a single day, the effects of daily resetting can cause leveraged and inverse funds to behave very differently from a simple multiple of the underlying index's longer-term return. They introduce complexity and risk beyond ordinary stock and bond funds and are not necessary building blocks for a beginner's long-term plan.

11.3 Options: Contracts, Calls and Puts

A contract with rights and obligations An option derives value from an underlying asset or index. A call holder has the right to buy; a put holder has the right to sell, at the strike price under the contract's exercise terms. A writer accepts the corresponding obligation if assigned. Buying a put does not mean borrowing stock to sell short. [S7]

The essential contract fields Identify underlying, call/put, strike, expiration, exercise style, settlement method, multiplier and deliverable. Standard U.S. equity options commonly cover 100 shares, but adjusted contracts after corporate actions may differ. Never assume every option delivers 100 shares. [S8]

Premium and contract cost Premium is commonly quoted per share for an equity option. An illustrative premium of $3.20 with a 100 multiplier costs $320 for one contract, plus applicable fees. Two contracts cost $640. Strike $50 implies $5,000 to purchase 100 shares on exercise of one standard call; the $320 premium is separate.

American and European styles American-style permits exercise before expiration; European-style restricts exercise to expiration under product terms. These labels describe exercise rights, not where a contract trades. Many U.S. stock/ETF options are American-style and share-settled; many index options are European-style and cash-settled. Check the specific product. [S7, S9]

Owning options versus owning shares An option holder generally does not receive the shareholder's voting rights or dividends merely by holding an option. Options have a finite life. An option can lose its entire premium even if the investor's directional view is eventually correct, because timing and price paid matter. Brokerage approval and the current OCC risk disclosure are part of learning the product. [S7]

11.4 Option Chains, Moneyness and Pricing

Chain fields An option chain groups calls and puts by expiration and strike. Read bid, ask, last trade, volume, open interest, implied volatility (IV) and Greeks. Volume counts contracts traded in the session; open interest counts outstanding contracts, generally published after processing. High open interest does not ensure a narrow spread or an easy exit. [S7, S8]

Intrinsic value and moneyness For stock price S and strike K: call intrinsic value = max(S - K, 0); put intrinsic value = max(K - S, 0). A call is in the money when S > K; a put when S < K. At the money means at or near the strike in common usage. Out-of-the-money options have zero intrinsic value. In the money does not automatically mean the trade is profitable. [S8]

Extrinsic value In ordinary quoted equity-option examples, extrinsic value is premium minus intrinsic value. A $50-strike call on a $53 stock priced at $4 has $3 intrinsic and $1 extrinsic value per share. Time, volatility, rates and dividends affect pricing; use the appropriate model and contract specifications. [S10]

Historical versus implied volatility Historical volatility measures past variation in returns over a stated window. IV is inferred from option prices using a pricing model; it is not a realized forecast or a prediction of direction. IV can differ across strikes and expirations. A volatility drop after earnings can reduce a long option's value despite a favorable stock move. [S10]

Execution example A fictional bid $3.00 / ask $3.30 has a $0.30 spread, or $30 per standard contract. A midpoint $3.15 is a reference, not a guaranteed fill. Last trade may be stale. Compare the contract's current quotes and available size; a limit order can control price but can remain unfilled.

11.5 Payoffs, Breakeven and Closing Trades

Long call at expiration Buy a $50-strike call for $3 per share, multiplier 100. Expiration profit = [max(S - $50, 0) - $3] × 100, excluding fees. At S = $48: -$300. At $52: -$100, despite being in the money. At $53: $0. At $60: +$700. Expiration breakeven is strike + premium = $53.

Long put at expiration Buy a $50-strike put for $2, multiplier 100. Profit = [max($50 - S, 0) - $2] × 100. At S = $55: -$200. At $49: -$100. At $48: $0. At $40: +$800. Expiration breakeven is strike - premium = $48.

Before expiration These breakevens apply at expiration. Before then, an option may be sold at a profit without the stock reaching the expiration breakeven, or at a loss even after a favorable move, because extrinsic value also changes. Closing-sale profit for a purchased option = (sale premium - purchase premium) × multiplier × contracts, less costs.

Opening and closing Buy to open establishes a long option; sell to close exits that long. Sell to open establishes a short option; buy to close exits the short. A closing trade must match the same contract series and position quantity. Selling to close an owned call is different from writing a new uncovered call. [S8]

Options loss versus resulting stock risk For an unexercised standalone purchased option, the premium paid plus costs is the loss limit. Exercise can create a stock position with additional funding needs and price exposure. Short options have different loss profiles: an uncovered call can have theoretically unlimited loss. Never apply the long-option premium limit to a writer or resulting stock position. [S7, S9]

11.6 Delta and Gamma

How to read all Greeks Greeks are model estimates of sensitivity, holding other inputs constant. They change with the market and do not guarantee fills, future returns or exact premium changes. This chapter uses per-share option values, a 100 contract multiplier and explicitly stated units. Platforms can display different scaling conventions. [S10]

Delta Delta estimates the option premium change for a $1 increase in the underlying. A long call usually has Delta between 0 and +1; a long put between -1 and 0. If a fictional call has Delta +0.60 and the stock rises $1,

its premium increases approximately $0.60, or $60 per standard contract, before other effects. Delta changes, so this is a local estimate. [S11]

Position Delta Two long standard calls with Delta +0.60 have position Delta +120: 2 × 100 × 0.60. For a small $1 stock rise, this implies approximately +$120. Two short calls reverse the sign. Delta can be used as a rough directional share equivalent, but options are not identical to owning that many shares.

Probability caveat Some traders use absolute Delta as a rough proxy for expiring in the money. It is not an exact real-world probability and is not the probability of making a profit. The model, dividends, volatility and the premium paid matter. [S11]

Gamma Gamma estimates the change in Delta for a $1 stock move. If Delta = 0.60 and Gamma = 0.04, a $1 rise changes Delta to about 0.64 locally. Long ordinary vanilla options generally have positive Gamma; short positions reverse it. Near-the-money options close to expiration can have high Gamma, making exposure change rapidly. [S12]

A curvature example Holding other inputs fixed, a local premium approximation is Delta × change in S + ½ × Gamma × (change in S)². With 0.60 Delta, 0.04 Gamma and a $2 rise: 0.60 × 2 + 0.5 × 0.04 × 4 = $1.28 per share. Large moves require repricing; the fixed-Greek estimate becomes less reliable.

11.7 Theta, Vega and Rho

Theta - passage of time Theta estimates premium change as time passes, often quoted per day. At Theta -$0.05 per share per day, one long standard contract loses about $5 of theoretical value over one day if other inputs remain constant. Ordinary long options generally have negative Theta; short options generally have positive Theta, with exceptions depending on contract economics. Confirm calendar-day versus trading-day convention. Decay is not linear; near-expiry, near-the-money options can decay quickly. [S13]

Vega - implied volatility Vega commonly measures premium change for a one-percentage-point IV increase. At Vega $0.12, IV rising from 25% to 27% implies about +$0.24 per share, or +$24 per standard long contract. That is a two-percentage-point change, not a 2% relative change. Ordinary long vanilla calls and puts generally have positive Vega; short positions reverse it. [S14]

Rho - interest rates Rho commonly measures premium change for a one-percentage-point rise in the model's interest-rate input. At call Rho +$0.08, rates rising from 4% to 5% imply about +$0.08 per share, or +$8 per standard long contract, all else equal. Ordinary long vanilla calls generally have positive Rho and puts negative Rho; short positions reverse the signs. Rate sensitivity often matters more for longer-dated options. [S15]

Why the stock direction is insufficient Suppose a long call gains an estimated $0.60 from a favorable stock move, loses $0.05 from one day of time and loses $0.72 from a six-point IV decline with Vega $0.12. The rough combined change is -$0.17 per share. A correct direction can still coincide with an option loss.

Unit checklist Delta: premium per $1 stock move. Gamma: Delta change per $1 stock move. Theta: premium per stated day. Vega: premium per stated IV unit. Rho: premium per stated rate unit. Multiply per-share exposures by signed contract count and multiplier. Do not multiply again if the screen already shows position-level Greeks.

11.8 Options Strategies, Exercise and Settlement Risks

Covered call Owning the underlying shares while writing a call can generate premium but limits upside above the strike when assigned. The shares can still lose substantial value. "Covered" describes delivery coverage, not protection from a falling stock. [S7]

Protective put Owning stock and buying a put can establish a strike-based sale right during the contract's life. The premium reduces returns. Coverage depends on matching shares, quantity and deliverable; protection expires. The combined trade can still lose money.

Cash-secured put Writing a put while reserving cash for assignment obligates the writer to buy shares at the strike if assigned. Premium does not eliminate downside: if the stock falls to zero, a standard position's expiration loss is approximately (strike - premium received) × multiplier, before costs.

Vertical spread Buying and selling options of the same type and expiration at different strikes creates a vertical spread. A fictional $50/$55 call debit spread bought for $2 has $200 maximum expiration loss and $300 maximum expiration gain for multiplier 100, assuming both legs are managed and settled as intended. Early assignment, leg closure, fees and a resulting stock position can create additional operational exposure. [S7]

Exercise and assignment Exercise uses the holder's contract right; assignment imposes the writer's obligation. American-style writers can be assigned before expiration. In-the-money short calls can face heightened early-assignment risk around an ex-dividend date. Selling an option to close can preserve remaining extrinsic value that exercise would give up. [S9]

Expiration and settlement Check the last trading time, exercise cutoff, automatic-exercise handling, settlement method and broker instructions. A $50 standard call can require $5,000 to buy shares. Near-strike expiration creates uncertainty about assignment and later stock exposure. AM versus PM settlement and overnight gaps can matter. Same-day-expiring options can change rapidly; knowing Greeks does not remove these risks. [S7, S9]

11.9 Fraud and Misleading Performance Claims

Investment fraud frequently features promises of high returns with little or no risk, pressure to act quickly before an "opportunity" disappears, unsolicited contact through social media or cold calls, and reluctance to provide clear written documentation or to allow verification of credentials. Past-performance claims that omit fees, selective time periods, or the risk of loss are incomplete and can be misleading. Official resources such as Investor.gov maintain checklists of common red flags and free tools to verify whether a person or firm is registered.

A useful habit is to ask: If this opportunity were as good and as safe as claimed, why is it being offered to me through an unsolicited channel rather than through established, regulated markets? Skepticism toward guaranteed or unusually high returns is a form of self-protection.

11.10 Verifying Financial Professionals

Before hiring anyone to manage money or give investment advice, check registration status and disciplinary history through Investor.gov's investment-professional search and FINRA's BrokerCheck. Confirm the person's credentials, understand the standard of care that applies (fiduciary versus suitability), and obtain important terms-services, fees, conflicts-in writing. Registration does not guarantee good performance or the absence of future problems, but the absence of required registration is a significant warning sign.

Common Misconceptions - Chapter 11

"Buying a call is the same as owning shares." Options have an expiration, separate pricing drivers and different rights.

"An in-the-money option guarantees profit." Premium and costs still matter.

"Greeks predict the exact next price." They are local model sensitivities, not guarantees.

"A covered call eliminates stock risk." The underlying shares can still fall substantially.

Chapter 11 Summary

Complex products add contractual obligations, leverage and financing risks. Calls and puts grant distinct rights to holders and obligations to writers. Premium, strike, expiration, multiplier, exercise style and settlement method determine the contract. Delta and Gamma describe stock-price exposure; Theta, Vega and Rho describe time, implied-volatility and rate sensitivities. Use stated units and signed contract counts. Investor protection also requires understanding product terms and verifying professionals rather than relying on performance promises.

Practice Questions - Chapter 11

1. Name two ways hedge funds typically differ from mutual funds in regulation or structure.

2. What is one reliable free resource for checking whether a financial professional is registered?

3. Why are leveraged and inverse ETFs generally considered inappropriate as core long-term holdings for beginning investors?

4. List two common red flags of investment fraud.

5. One standard call costs $2.40 per share. What premium is paid, excluding fees?

6. A $100-strike call bought for $4 expires with the stock at $102. What is expiration profit for multiplier 100?

7. Two long calls each have Delta 0.45 and multiplier 100. What is position Delta?

8. Gamma is 0.03 and Delta is 0.45. Estimate Delta after a $1 stock rise.

9. Vega is $0.10 per share per IV point. IV falls from 30% to 27%. Estimate the value change for one standard long option.

10. Theta is -$0.04 per share per day. Estimate one-day decay for three standard long contracts.

11. Rho is -$0.06 per share per rate point. Rates rise from 3% to 4%. Estimate one standard long option’s change.

12. Is “in the money” the same as profitable?

Student Case Studies

The following case studies are hypothetical. They illustrate how the concepts in earlier chapters can be applied to realistic student situations. They are not recommendations for any particular reader.

Case Study A - Building the First Emergency Fund

Jordan is 19, works part-time during the semester, and has about $800 in a checking account that fluctuates with rent and food expenses. After reading Chapter 1, Jordan opens a separate high-yield savings account at an FDIC-insured bank and sets up an automatic $50 transfer every payday. Within roughly a year the emergency fund reaches about $1,400. Jordan's next goal is three months of essential expenses. The automatic transfer continues while Jordan also begins contributing enough to a new employer's 401(k) to capture the full available match.

Discussion points: Why keep the emergency fund in a separate account rather than in checking? Why capture the employer match even while the emergency fund is still growing? What would change if Jordan had high-interest credit-card debt?

Case Study B - Opening a First Investment Account

Sam is 22, has a stable full-time job, a completed three-month emergency fund, and no high-interest consumer debt. Sam opens a Roth IRA and begins automatic monthly contributions of $100 into a low-cost total-U.S.-stock-market index fund. Sam chooses the Roth structure because current taxable income is relatively low and is expected to rise with career progression. Before investing, Sam reads the fund's summary prospectus, notes the expense ratio, and confirms that the fund seeks to track a broad market index. Sam understands that the account value will fluctuate and has no plan to withdraw the money for at least a decade.

Discussion points: Why is the emergency fund treated as a prerequisite? What does "low-cost" mean in practice, and where is that information found? Why might a Roth IRA be attractive to someone early in a career?

Case Study C - Managing a Concentrated Position

Alex received shares of a single large technology company as a gift several years ago and watched the value rise substantially. The position now represents more than half of Alex's total financial assets. After studying diversification in Chapter 8, Alex decides to sell a portion of the shares gradually over time (paying attention to tax consequences in a taxable account) and to redirect the proceeds into a broader total-market index fund. The process is deliberately gradual to manage both taxes and the emotional difficulty of selling a holding that has performed well.

Discussion points: What risks does concentration create even when the concentrated holding has performed strongly in the past? How can spreading sales over time help with both taxes and behavior? What role does a written plan play in following through?

Case Study D - Balancing Near-Term and Long-Term Goals

Taylor is 24 and wants both to save for a possible house down payment in about five years and to begin retirement saving. After building a modest emergency fund, Taylor splits new savings: a portion goes into a conservative mix of short-term bond funds and high-yield savings earmarked for the housing goal, and a portion goes into a stock-heavy target-date or total-market fund inside a Roth IRA for retirement. Taylor revisits the split once a year and will shift the housing-goal money toward cash as the purchase date approaches.

Discussion points: Why use different vehicles for goals with different time horizons? What happens to the housing-goal allocation as the five-year mark nears? How does the existence of the emergency fund affect the risk that can be taken with the other two goals?

Answer Key to Practice Questions

Chapter 1

1. An asset is something owned that has value (example: cash in a savings account, a used car). A liability is something owed (example: student-loan balance, credit-card debt).

2. An emergency fund provides cash for unexpected necessary expenses so the investor is not forced to sell investments at a possible loss or to rely on high-interest debt.

3. Three months × $765 = $2,295.

4. Avalanche: lowest total interest cost. Snowball: faster psychological wins from eliminating small balances.

5. A one-month fund is more achievable and still provides meaningful protection; it builds the habit and can be expanded later.

Chapter 2

1. Compound interest earns returns on previously earned interest, so the base grows; simple interest earns returns only on the original principal.

2. 1 / (1.03)^20 is about 0.554, so roughly 44.6 percent of purchasing power is lost; about $554 of today's purchasing power remains.

3. Because compounding has more periods in which to operate; later starts require much larger monthly amounts to reach the same goal.

4. Nominal return is the stated percentage change; real return adjusts for inflation and reflects the change in purchasing power.

5. The fee difference compounds against the investor every year; over decades the cumulative gap in ending wealth becomes large even if gross returns are similar.

Chapter 3

1. Federal deposit insurance (up to applicable limits) and high liquidity / ready access without market-price fluctuation.

2. FDIC protects depositors against the failure of an insured bank up to the coverage limit. SIPC protects customers of member broker-dealers against the failure of the firm (missing securities or cash) up to its limits, but does not protect against market losses.

3. Money market funds are securities, not bank deposits; they aim for stability but are not backed by FDIC insurance.

4. Liquidity is the ease of converting an asset to cash without significant loss of value. A stock fund can be sold quickly but at an uncertain price; a savings account offers both speed and price stability for a two-year goal.

Chapter 3A

1. $1,500 × 0.04 × 9/12 = $45 interest; ending value $1,545.

2. Monthly rate = 0.06/12 = 0.005. Balance = $1,000(1.005)^12 = $1,061.68; APY ≈ 6.168%. APY already incorporates compounding.

3. $50 × 12 = $600 contributions. Balance = $50[(1.003)^12 - 1]/0.003 = $610.00; interest ≈ $10.00.

4. $7 × 12 = $84.

5. $10 at 0.5% APY versus $60 at 3% APY; $50 gross difference. Compare fees, minimums, rate variability and access conditions.

6. An MMDA is an eligible bank deposit at an insured bank, subject to coverage limits. A money market mutual fund is a security without FDIC deposit insurance.

7. $2,500 × 1.04 = $2,600. Check the CD agreement, penalty provision, maturity/renewal notice and grace-period rules.

8. $1,000 per rung; maturities at years 1, 2 and 3. Unmatured rungs may incur penalties, and renewal rates are unknown.

9. $40 × 10 - $12 = $388.

10. Check the initial waiting period and whether withdrawal must close the entire CD; also verify fees and other restrictions.

11. Total = $265,000; $15,000 above the assumed $250,000 single-owner coverage limit.

12. No. A recurring transfer usually automates contributions into an account; a fixed rate or maturity requires separate product terms.

Chapter 3B

1. ($600 + $200) ÷ $4,000 = 20%. No: approval depends on the full application and product rules.

2. Fee = $150. Cash received = $4,850; principal owed = $5,000.

3. Base borrowing = $18,000; LTV = 90%. Check credit, income/debts, vehicle criteria, title and required insurance.

4. Private loans use lender credit/repayment checks and institution-specific terms; federal Direct loans use federal-aid eligibility and offer distinct protections. A co-signer shares responsibility for repayment and can suffer credit harm.

5. Base loan = $240,000; LTV = 80%. Taxes, insurance, any mortgage insurance, association fees and maintenance add to housing costs.

6. VA adds service-based entitlement/COE and lender/property conditions. USDA adds eligible location and household-income/program rules. FHA low down payments still require eligible borrowers and properties and underwriting.

7. $300,000 × 0.80 - $180,000 = $60,000, subject to all other lender conditions.

8. Under-21 card applicants generally need independent ability to pay or an eligible person aged 21 or older sharing liability who can pay; issuer offerings vary. Simplified monthly line interest = $500 × 0.12/12 = $5.

9. $52 × 12 - $600 = $24. Credit-builder funds are commonly held until contractual release conditions are satisfied.

10. $1,500 ÷ $1,000 = 1.5. No: business eligibility, ownership/residency, repayment evidence and other program conditions also apply.

11. Savings = $40 × 12 = $480; net result = $480 - $600 = -$120. Private refinancing of federal loans gives up federal benefits and protections.

12. Value = 5 × $32 = $160; loss = $40, or 20% of $200. Bank branding does not provide FDIC insurance for securities.

13. Six-month coupon = $25. Simplified result = $950 + $25 - $1,000 = -$25 before costs.

14. Gain = $20; holding-period return = $20 ÷ $980 ≈ 2.04%. Annual yield requires the holding period and calculation convention.

15. Mutual-fund shortfall = $300. Three whole ETF shares cost $240, leaving $10 before costs.

16. Half the index gain is 10%, reduced to 6% by the cap. Return = $60, with $1,060 total at maturity if conditions are met. Issuer default, contract conditions and early-sale risks remain.

17. Permitted capacity = $70,000 × 0.50 = $35,000. Excess debt = $40,000 - $35,000 = $5,000.

18. For a loan, collect identity, repayment and debt evidence; obtain product criteria, APR/fee disclosures, payment schedule and collateral terms. For securities, verify the legal brokerage, account requirements, product minimums, prospectus/offering documents, fees, permissions and insurance/protection disclosures.

Chapter 4

1. A stockholder owns a residual ownership interest in the company; a bondholder owns a creditor claim (a loan) that ranks senior to equity in bankruptcy.

2. Market capitalization is share price times shares outstanding. In market-cap-weighted indexes, larger companies receive greater weight and therefore influence index performance more.

3. Interest-rate risk and credit (default) risk are two central bond risks; inflation risk on fixed coupons is another.

4. Because rising market interest rates cause existing bonds with lower coupons to decline in market price; the investor who sells before maturity realizes the loss even if the issuer never misses a payment.

5. Market cap: $80 × 50 million = $4 billion. P/E: $80 ÷ $4 = 20×. Dividend yield: $1.60 ÷ $80 × 100 = 2%.

6. No. Today may be incomplete, and the average covers complete sessions.

7. No. Broker inventory is estimated availability through that broker; short interest is outstanding shorts at a reporting date.

8. $8,000 × 0.09 × 5/360 = $10.

Chapter 5

1. NAV is the fund’s assets minus liabilities, divided by shares outstanding-the per-share value of the portfolio.

2. Mutual-fund shares are bought/redeemed at end-of-day NAV; ETF shares trade throughout the day on an exchange at market prices.

3. The higher fee compounds against the investor every year; over decades the cumulative difference in ending wealth can be substantial.

4. In the fund’s prospectus (especially the fee table and principal-risks section) and related shareholder reports.

Chapter 6

1. No. An index is a statistical measure; investors buy funds or other products that seek to track the index.

2. Lower costs (expense ratios and usually lower trading costs), which improve the probability of higher net returns relative to higher-cost active funds over long periods.

3. Tracking error is the difference between the fund’s performance and the index’s performance; investors care because large or persistent tracking error means the fund is not delivering the intended exposure.

Chapter 7

1. No. The Nasdaq-100 deliberately excludes financial companies.

2. The number labeled “NASDAQ” is typically the Composite (3,000+ stocks including financials); most Nasdaq ETFs track the Nasdaq-100 (100 large non-financial stocks).

3. A large IPO enters the Composite quickly at full market-cap weight; entry into the Nasdaq-100 is subject to eligibility rules, timing of reconstitution/rebalance, and concentration constraints that limit its weight.

Chapter 8

1. Risk tolerance is the emotional willingness to accept volatility; risk capacity is the financial ability to absorb losses without harming essential goals.

2. Company-specific events affect a smaller percentage of a 30-stock portfolio than of a 3-stock portfolio, reducing the impact of any single failure.

3. Rebalancing restores the target asset allocation after market movements cause drift; it controls risk and enforces a disciplined “sell high, buy low” pattern.

4. Automatic monthly investing purchases more shares when prices are lower and fewer when prices are higher, which is the definition of dollar-cost averaging, without requiring market forecasts.

Chapter 9

1. Investor.gov’s investment-professional search tool and FINRA’s BrokerCheck.

2. To disclose the fund’s investment objectives, principal risks, fees and expenses, and other material information so investors can make informed decisions.

3. A market order prioritizes immediate execution at the current market price; a limit order prioritizes a specified price (or better) and may not execute.

4. Compensation structure can create conflicts of interest; understanding it helps the investor evaluate whether advice is aligned with the investor’s interests.

Chapter 10

1. The match is additional compensation that is otherwise left unclaimed; it provides an immediate return equal to the match rate on the contributed dollars.

2. Traditional contributions are pre-tax (reduce current taxable income) and withdrawals are taxed as ordinary income; Roth contributions are after-tax and qualified withdrawals are tax-free.

3. Longevity risk is the risk of outliving one’s savings; it matters because retirement can last decades and running out of money is a primary failure mode of retirement plans.

4. RMD ages and calculation rules have changed in recent legislation and may change again; relying on outdated figures can produce incorrect planning or tax penalties.

Chapter 11

1. Hedge funds are generally limited to investors meeting financial/sophistication tests; they face fewer regulatory constraints on leverage, liquidity, and disclosure than mutual funds.

2. Investor.gov or FINRA BrokerCheck.

3. They are designed for short-term trading; daily resetting can cause performance over longer periods to differ substantially from a simple multiple of the index, adding complexity and risk unsuitable for most beginners' core holdings.

4. Promises of high returns with little or no risk; pressure to act quickly; unsolicited offers; reluctance to provide written documentation or allow credential verification.

5. $2.40 × 100 = $240.

6. [$2 intrinsic - $4 paid] × 100 = -$200.

7. 2 × 100 × 0.45 = +90.

8. Approximately 0.48, other inputs fixed.

9. 0.10 × (-3) × 100 = -$30.

10. -$0.04 × 3 × 100 = -$12.

11. -$0.06 × 1 × 100 = -$6.

12. No. Moneyness compares underlying price with strike. Profit also reflects premium paid, costs and when the position is closed.

Glossary

52-week high / low

Highest and lowest prices over a provider’s rolling 52-week window; session and adjustment conventions matter.

American-style option

An option that permits exercise before expiration under its contract terms.

Amortization

Repayment of a loan over time through scheduled amounts allocated to interest and principal.

Annual percentage rate (APR)

An annualized borrowing-cost measure including interest and specified charges under the applicable disclosure rules; distinct from deposit APY.

Annual percentage yield (APY)

An annualized deposit yield that includes compounding under the disclosed assumptions; variable rates may change.

Ask

A quoted price at which a seller is offering a security.

Asset

Something owned that has economic value.

Asset allocation

The division of a portfolio among major asset classes such as stocks, bonds, and cash.

Assignment

Notification that an option writer must fulfill the contract obligation.

At the money

An option whose underlying price is at or near its strike price.

Automatic savings plan

Scheduled recurring transfers or deposits used to build savings; not itself a promise of a fixed rate or maturity.

Average volume

Mean trading volume over a specified lookback period.

Bid

A quoted price at which a buyer is offering to purchase a security.

Bond

A debt security representing a loan from the investor to the issuer.

Borrow rate

A commonly annualized fee rate for borrowing securities; the basis and day count depend on the broker.

Call option

A contract giving its holder the right to buy the specified underlying or receive a contractual cash settlement.

Capital gain

Profit realized when an investment is sold for more than its purchase price.

CD ladder

A set of CDs with staggered maturity dates to provide scheduled access to portions of savings.

Certificate of deposit (CD)

A time deposit with a stated term and contractual interest and withdrawal conditions.

Checking account

A deposit account designed primarily for frequent payments and withdrawals.

Club savings account

A goal-oriented deposit plan that may specify regular deposits, a payout date and restrictions on early access.

Co signer

A person who shares responsibility for repayment under a credit agreement; more than a personal reference.

Collateral

An asset pledged to secure repayment of a loan; default can lead to seizure or sale under applicable terms and law.

Combined loan to value (CLTV)

Total debt secured by a property divided by its value, expressed as a percentage.

Compound interest / compound growth

Earnings generated on both principal and previously accumulated earnings.

Contract multiplier

The factor used to scale a quoted option value into contract value; commonly 100 for standard U.S. equity options.

Credit builder loan

A loan commonly holding proceeds in a restricted account until repayment/release conditions are met, intended to support a payment record.

Debt service coverage ratio (DSCR)

Cash available for debt service divided by required debt service over the same period; lender calculation methods vary.

Debt to income ratio (DTI)

Required monthly debt payments divided by gross monthly income; definitions and qualifying thresholds vary.

Delta

Local sensitivity of option premium to a $1 increase in the underlying price.

Diversification

Spreading investments across and within asset classes to reduce the impact of any single security or sector.

Dividend

A distribution of a portion of a company’s earnings to shareholders.

Dividend yield

Annual dividend per share divided by share price, expressed as a percentage; the annual dividend basis must be specified.

Dollar-cost averaging

Investing fixed amounts at regular intervals, which purchases more shares when prices are lower and fewer when prices are higher.

Early withdrawal penalty

A contractual charge for taking funds from a time deposit before maturity; it can sometimes reduce principal.

Earnings per share (EPS)

A company’s reported earnings attributable to each share on the stated accounting and share-count basis.

Emergency fund

Cash reserves set aside for unexpected necessary expenses.

European-style option

An option whose exercise is restricted to expiration under the product terms.

Exchange-traded fund (ETF)

An investment fund that trades on an exchange throughout the day and typically holds a portfolio of securities.

Exercise

Use of an option holder’s contractual right to buy, sell or receive cash settlement.

Expense ratio

Annual fund operating expenses expressed as a percentage of assets under management.

Expiration

The end of an option’s contractual life, subject to product trading and exercise deadlines.

Extrinsic value

In ordinary equity-option examples, the premium beyond intrinsic value.

FDIC insurance

Federal deposit insurance protecting bank depositors up to applicable limits against the failure of an insured bank.

Fiduciary

A person or firm required to act in the client’s best interest.

Fixed rate loan

A loan whose stated interest rate remains unchanged for the specified fixed-rate period.

Gamma

Local sensitivity of Delta to a $1 increase in the underlying price.

Greeks

Model-based option sensitivities to changes in pricing inputs, holding other inputs constant.

High today / low today

Highest and lowest reported trade prices so far in the specified session.

High yield savings account (HYSA)

A savings deposit marketed for a comparatively competitive yield; the label does not create a separate insurance category.

Historical volatility

A measure of past variability in returns over a specified period.

Home equity

The difference between a home’s value and debt secured by it; not necessarily cash available to borrow.

Home equity line of credit (HELOC)

A revolving credit line secured by a home, with contractual draw and repayment periods.

Implied volatility (IV)

Volatility inferred from an option price using a pricing model; it does not predict direction.

In the money

A call with underlying above strike, or a put with underlying below strike; not a guarantee of profit.

Index

A statistical measure of the performance of a defined group of securities.

Index fund

A mutual fund or ETF that seeks to track the performance of a market index.

Inflation

A general rise in the prices of goods and services over time, reducing purchasing power.

Installment loan

A loan repaid through scheduled payments over a term.

Interest rate

The stated rate used to calculate interest; unlike APY, a nominal annual rate does not necessarily incorporate compounding.

Intrinsic value

For an ordinary stock call, max(stock price minus strike, zero); for a put, max(strike minus stock price, zero).

Liability

Something owed; a debt.

Liquidity

The ease with which an asset can be converted to cash without significant loss of value.

Loan Estimate

A standardized mortgage disclosure used to compare estimated loan terms, payments and closing costs for covered transactions.

Loan to value ratio (LTV)

Loan balance divided by collateral value, usually expressed as a percentage.

Market capitalization

Share price multiplied by the number of shares outstanding.

Minimum balance

A required balance whose purpose and measurement may differ for opening an account, earning interest or waiving fees.

Money market deposit account (MMDA)

A bank deposit that may offer interest and limited transaction features; distinct from a money market mutual fund.

Money market mutual fund

A security investing in short-term debt instruments; it is not an FDIC-insured bank deposit.

Mortgage

A loan secured by real property, with foreclosure risk if the borrower fails to meet obligations.

Mutual fund

An open-end investment company that pools investor money and issues redeemable shares priced at NAV.

Net asset value (NAV)

The per-share value of a fund’s assets minus its liabilities.

Net worth

Total assets minus total liabilities.

Open interest

The number of outstanding option contracts for the specified series or market at the reporting time.

Open price

The opening price for the trading session defined by the data provider.

Option chain

A display of option series and quotes organized by strikes and expirations.

Origination fee

A charge for arranging or making a loan; payment method and APR treatment depend on the loan and applicable rules.

Out of the money

An option with no intrinsic value at the given underlying price.

Premium

The option price, commonly quoted per share for equity options and scaled by the contract multiplier.

Price-earnings ratio (P/E)

Share price divided by annual EPS; specify trailing or forward earnings and treat losses or zero earnings carefully.

Prospectus

The legal disclosure document for a mutual fund, ETF, or security offering.

Put option

A contract giving its holder the right to sell the specified underlying or receive a contractual cash settlement.

Rebalancing

Adjusting portfolio holdings to restore a target asset allocation after market movements cause drift.

Recurring deposit plan

Regular periodic contributions; in U.S. banking usually an automatic savings plan or recurring transfer unless a distinct contract specifies otherwise.

Refinancing

Replacing existing debt with a new loan, potentially changing rates, term, costs, collateral or protections.

Revolving credit

Credit that may be borrowed, repaid and borrowed again within contractual limits.

Rho

Option premium sensitivity to the interest-rate input, commonly per one-percentage-point increase.

Risk capacity

The financial ability to absorb investment losses without jeopardizing essential goals.

Risk tolerance

The emotional and psychological willingness to accept investment fluctuations.

Savings account

An interest-bearing deposit used to set aside money, with access and fee conditions determined by the institution.

Secured loan

A loan backed by pledged collateral; collateral does not remove the borrower’s repayment obligation.

Securities backed line of credit (SBLOC)

Borrowing secured by eligible investments, subject to lending limits, collateral calls and potential forced sales.

Short interest

Outstanding short positions at a stated reporting date; distinct from broker borrow availability.

Short inventory

A platform-dependent term commonly referring to estimated shares available to borrow through a broker.

Short selling

Borrowing securities to sell and later buying securities to return; stock-price rises can cause large losses.

Simple interest

Interest calculated on principal alone: principal × annual rate × time in years.

SIPC

Securities Investor Protection Corporation; provides limited protection to customers of member broker-dealers against firm failure (not against market loss).

Spread (bid-ask)

The difference between quoted ask and bid prices.

Stock (equity)

A security representing an ownership interest in a corporation.

Strike price

The contractual exercise price of an option.

Structured note

An issuer’s debt security with a payoff linked to reference assets and contractual conditions; distinct from an insured deposit.

Theta

Option premium sensitivity to passage of time, commonly expressed per stated day.

Time horizon

The length of time an investor expects to hold an investment or to need the money for a goal.

Total return

The combination of price appreciation (or depreciation) and income (dividends or interest) over a period.

Unsecured loan

A loan without specifically pledged collateral; default may still cause credit damage and collection.

Variable rate

An interest rate that may change under the account terms.

Vega

Option premium sensitivity to implied volatility, commonly per one-percentage-point IV increase.

Volume

The number of shares or option contracts traded during a specified period.

Source Bibliography

This independent teaching handbook consolidates concepts from the primary educational resources below. Numbered source IDs in Chapters 4 and 11 identify the relevant detailed references. Numerical examples are hypothetical. Sources for stock metrics and options were reviewed October 4, 2026; consult live documents for changing rules, product features and broker-specific conventions.

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[S2] Investor.gov: Price-Earnings Ratio https://www.investor.gov/introduction-investing/investing-basics/glossary/price-earnings-pe-ratio

[S3] Fidelity: Stock Quote Field Definitions https://www.fidelity.com/webxpress/help/topics/help_quote_stock.shtml

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Investor.gov - Characteristics of Mutual Funds and ETFs (Investor Bulletin) https://www.investor.gov/introduction-investing/general-resources/news-alerts/alerts-bulletins/characteristics-mutual-funds-exchange-traded-funds

Investor.gov - Hedge Funds https://www.investor.gov/introduction-investing/investing-basics/investment-products/private-investment-funds/hedge-funds

Investor.gov - Bonds FAQs https://www.investor.gov/introduction-investing/investing-basics/investment-products/bonds-or-fixed-income-products/bonds

Investor.gov - Beginners' Guide to Asset Allocation, Diversification, and Rebalancing https://www.investor.gov/additional-resources/general-resources/publications-research/info-sheets/beginners-guide-asset

Investor.gov - Glossary (various terms) https://www.investor.gov/introduction-investing/investing-basics/glossary

Investor.gov - Retirement Toolkit and related account pages https://www.investor.gov/additional-resources/retirement-toolkit

FINRA - Investing Basics https://www.finra.org/investors/investing/investing-basics

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Nasdaq - Nasdaq Composite vs. Nasdaq-100: What Investors Should Know https://www.nasdaq.com/newsroom/nasdaq-composite-vs-nasdaq-100-what-investors-should-know

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FDIC - Deposit Insurance https://www.fdic.gov/resources/deposit-insurance

FDIC - Financial Products That Are Not Insured by the FDIC https://www.fdic.gov/resources/deposit-insurance/financial-products-not-insured

This handbook is not an official publication of the SEC, FINRA, Nasdaq, FDIC or OCC. Organization names identify source provenance; their inclusion does not imply endorsement. Verify changing rules and contractual terms with the responsible source.

Banking Sources Added in This Edition

Banking sources reviewed October 12, 2026. These references explain product concepts and disclosures; example rates and terms are invented for teaching. Verify the institution's current contract and official guidance. FDIC and CFPB are additional source organizations, without endorsement of this handbook.

[B1] FDIC Deposit Insurance FAQs

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[B2] FDIC Your Insured Deposits

https://www.fdic.gov/resources/deposit-insurance/brochures/insured-deposits

[B3] FDIC Consumer News September 2018 Savings and Money Market Accounts

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[B4] CFPB Regulation DD Account Disclosures

https://www.consumerfinance.gov/rules-policy/regulations/1030/4/

[B5] CFPB Regulation DD Definitions

https://www.consumerfinance.gov/rules-policy/regulations/1030/2/

[B6] CFPB Certificate of Deposit Rollover or Renewal

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[B7] CFPB CD Rates and Product Options

https://www.consumerfinance.gov/ask-cfpb/the-interest-rate-offered-for-cds-certificates-of-deposit-is-low-is-there-anything-i-can-do-about-that-en-921/

[B8] CFPB Interest Paying Checking Accounts

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Coverage verification tools: FDIC BankFind https://banks.data.fdic.gov/bankfind-suite/bankfind and FDIC EDIE https://edie.fdic.gov/ . Credit union coverage: NCUA https://ncua.gov/consumers/share-insurance-coverage .

Loan and Securities Sources Added in This Edition

Sources reviewed October 12, 2026. Eligibility summaries describe common requirements and current program distinctions, not a complete underwriting checklist. Confirm current conditions with the lender, school or official agency. All rates, payment scenarios and illustrative cutoffs are hypothetical. Securities references extend the existing investment chapters without implying endorsement by any organization.

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[L2] CFPB Auto Loan Rate and Underwriting Factors

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[L3] CFPB Ready to Buy a Home

https://www.consumerfinance.gov/consumer-tools/mortgages/ready-to-buy-a-home/

[L3] CFPB Home Affordability and Full Housing Costs

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[L4] CFPB Private Student Loans

https://www.consumerfinance.gov/ask-cfpb/what-are-private-student-loans-en-2136/

[L4] CFPB Student Loan Co Signers

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[L4] CFPB Consolidating or Refinancing Student Loans

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[L5] CFPB Home Equity Loans and HELOCs

https://www.consumerfinance.gov/ask-cfpb/what-is-the-difference-between-a-home-equity-loan-and-a-home-equity-line-of-credit-heloc-en-247/

[L5] CFPB HELOC Consumer Brochure

https://files.consumerfinance.gov/f/documents/cfpb_heloc-brochure.pdf

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[L8] Federal Student Aid General Eligibility

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[L8] Federal Student Aid Subsidized and Unsubsidized Loans

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[L11] VA Home Loan Eligibility

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[L12] USDA Single Family Housing Guaranteed Loan Program

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[L13] SBA 7a Loans

https://www.sba.gov/loans/7a-loans/

[L14] SBA Procedural Notice 5000 876626 Ownership Citizenship and Residency Effective March 1 2026

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https://www.investor.gov/introduction-investing/investing-basics/investment-products/mutual-funds-and-exchange-traded-2

[L18] TreasuryDirect Marketable Securities FAQs

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[L19] FINRA Securities Backed Lines of Credit

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[L20] FDIC Financial Products Not Insured

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[L20] FDIC Your Insured Deposits

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